



**KTH Industrial Engineering
and Management**

Strategic energy systems analysis:
Possible pathways for the transition of electricity sector in
Tanzania

MJ210X - Degree Project in Energy Systems Analysis
Master Thesis Report

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Possible pathways for the transition of electricity
sector in Tanzania**

Georgios Avgerinopoulos

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31 December 2013	Mark Howells	Mark Howells
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Abstract

This study examines the concept of the evolution of electricity sector in Tanzania. Electrification of Africa has raised large discussion and thus, nine scenarios based on different production pathways and demand projections are formulated. The study considers both grid based centralized electricity and decentralized power production. The main differentiation is between a centralized electricity system and decentralized power that are closer to demand. A model is created using three modeling tools (Answer-OSeMOSYS, LEAP and MESSAGE) and the results are presented and compared. Finally, different funding options for electricity expansion projects in Tanzania are explored in order to investigate the feasibility of the scenarios as well as a geopolitical analysis is carried out.

Keywords: Tanzania, electrification, electricity expansion, energy system, centralized system, decentralized power plants, Answer-OSeMOSYS, LEAP, MESSAGE

Acknowledgments

Accomplishing this challenging task, I would like to thank from the bottom of my heart first of all my brother Niko, my parents Sissy and Grigori and my super cool friends Madlen and Sotiri for tolerating my absurd personality all these years and helping me discover myself and my skills.

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This work is dedicated to all those radical minds (scientists or not) from ancient times to present, who have shown that only by thinking outside the box you can change this mysterious world.

Georgios

Stockholm, 2013

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1. Introduction

In part, stemming from the Millennium Development Goals set by the UN, electrification of developing countries has opened a discussion involving several groups of experts and stakeholders (UN, 2013). As specific action is the Secretary General's 'Sustainable Energy for All' initiative (SE4A) (Sustainable-Energy-for-All, 2013). In Africa, the rate of access to electricity is no more than 64% for urban households and 13% for rural ones (IEA/OECD, 2012). Apart from low electrification rates, the quality of the electricity is poor. This poor quality electrification in Africa impacts harmfully its economic and social development (Foster, 2011). This is one of the reasons why the African access to electricity is discussed worldwide. As electricity is needed for social development (providing better access to basic needs such as education) as well as economic development, providing Africa with a better electricity grid is part of the global sustainable development.

The Tanzanian case illustrates well the sub-Saharan African situation. In Tanzania, the access to electricity is no more than 14% (2009) (World-Bank, 2010). Moreover, the demand for electricity in this country constantly increases due to the current 6,5% GDP growth rate (African-Economic-Outlook, 2013). Further, as electricity investments are often long lived quick and clever decision making is needed in order to build an appropriate electricity sector in Tanzania. Unfortunately in the face of investment uncertainty as well as their relatively fast deployment times, gas turbines (running on oil or natural gas) are often the investment of choice.

Tanzania as well as many African countries has significant potential in renewable sources and this implies access to fuel-free energy sources (JRC, 2011). However, the needed technologies to exploit these sources (sun, wind, biomass and geothermal for Tanzania) require effort including economic cost benefit analysis, strong institutions as well as funding. In this thesis the focus is on least cost optimization planning to determine the most economic electricity system as well as a scan of potential funding options.

Increasing electricity supply in Africa has raised large debate on the scale of the technologies that should be installed and the way these should be managed. Some actors support the grid-connected, centralized power plants, controlled by the government or private utility companies while some groups of experts claim that the best solution for Africa is to install small, off-grid power plants controlled by the local communities. (Sebitosi, 2010)

An energy system consists of a combination of several components which interact in a dynamic way. Thus, managing this complexity is a demanding important task – compounded by the need to take into account how the system operates now and how it might evolve in the future. Long term energy system models aim help abstract important attributes of the system's components. By quantifying the way these components interact provide models help provide consistent results. Further, optimization models show how 'best' to arrange the components for a given goal. By

‘components’, we typically mean technologies such as power plants or transmission systems, fuels and emissions. Energy models constitute a very useful tool for policy makers but they are not enough. There are for instance social and geopolitical aspects that must be taken into account which cannot be quantified. However, a well structured model can lay the foundations for the design of and insights into energy policies and strategies.

1.1 Research question and objectives

This study aims to answer the following research question: *Can one identify indicatively optimal expansion pathways for the transition of electricity sector in Tanzania?*

The research is performed based on the following objectives. First, nine different scenarios are formulated and a model is created (using three different tools) to examine their results. In addition, these results are compared in order to shed light on each scenario. Further, as the tools that are employed are similar in nature differences (if any) between their outputs are noted. Finally, selected renewable energy funding opportunities for projects in Tanzania are investigated as well as some geopolitical aspects.

1.2 Scope

The current study deals with the electricity sector of Tanzania and its evolution. Currently, Tanzania relies mainly on hydropower and some thermal power stations (natural gas fired). Apart from the existing plants, there are some others that are planned to be built by the year of 2025 (SNC-LAVALIN-INTERNATIONAL-INC, 2011). Regarding the demand for electricity, three projections are considered (low, base, high) based on economic growth expectations.

1.3 Limitations

The research is narrowed by a number of limitation. Some of these include:

- The demand for electricity is not divided into sub-sectors (households, services, industry etc). This leads to a simplified demand profile and the contribution of each sub-sector cannot be discussed.
- The electricity sector is studied without taking into account the interactions with other energy sectors such as biomass for cooking or heating. A holistic approach could give different results. For instance, natural gas might not be the most cost effective option for electricity but a combined heat and power plant would run at a significantly lower cost (Howells M., 2002).

- The impact of the evolution of the electricity sector on climate, land and water is not investigated. Taking all these factors into account could either impose some constraints (i.e. conflict between hydropower and fresh water withdrawal) or provide better solutions such as use of agricultural waste instead of crops for fuel production which would reduce water consumption and increase food security (Welsch, 2014)
- We simplify short term power plant dynamics (not including detailed demand profiles or issues surrounding the intermittency of renewables). This can have an impact on the phenomenal performance of different technologies and make for example renewables seem more competitive than they actually are (Welsch, n.d.).

1.4 Methodology

As a starting point a literature review is undertaken. It focuses both on scientific papers and publically available information, in particular international institutions' with a focus on supporting development. Next step is the data collection. A key aspect of the investigation is to attempt to find or estimate realistic data. These include technology and fuel costs, historical and planned investments in the sector as well as energy demand projections. Finally, a model is created in Answer-OSeMOSYS, LEAP and MESSAGE based on the available data and assumptions and scenarios are developed.

1.5 Tool description

In general, multiple types of tools are used to address different cases. These tools are mainly classified in the following categories: simulation, accounting and optimization. This study focuses on optimization. This type of tool aims at giving solution to a problem at the highest or lowest value for a particular factor. A common case (like the current model) is the least cost optimization where the target is to meet the demand through the entire modeling period at the lowest possible cost (Welsch, 2013).

Different institutions tend to use various pieces of energy modeling software. Each of those tools has its strengths and weaknesses. Below, a short description of the tools used in this project is given. It should be noted that these tools are continuously be developed, and that the tool related observations in this thesis are based on the latest available versions.

OSeMOSYS stands for Open Source Energy Modeling System and is an optimization tool. It is created by a group of leading institutions and is suitable for academic investigations, powerful energy systems analysis and prototyping new energy model formulations. It is written in the free and open source software GNU Mathprog which is a mathematical programming language. Different interfaces can be used to run the

OSeMOSYS code. In this case the chosen piece of software is Answer-OSeMOSYS¹ (OSeMOSYS, 2013).

LEAP² stands for Long range Energy Alternatives Planning System and is an accounting tool. It is commonly used for climate change mitigation assessment and is characterized by a high level of flexibility. An optimization plug-in has been recently developed (COMMEND, 2013).

MESSAGE³ stands for Model for Energy Supply Strategy Alternatives and their General Environmental Impacts and is an optimization tool. It is built by the International Institute for Applied Systems Analysis (IIASA). The tool was recently acquired by the IAEA where some certain improvements were made in it, regarding mainly its user-interface (IAEA, 2007).

2. Data collection

2.1 Demand

The electricity demand in Tanzania is growing rapidly, with a yearly growth rate is close to 5%, compared to Sweden which is at 0.46% (IEA, 2013). Demand growth (including peak demand) from 2008 to 2038 is taken from the *Regional power system master plan and grid code study* for Eastern Africa power pool and East African community (SNC-LAVALIN-INTERNATIONAL-INC, 2011), written in association with PARSONS BRINCKERHOFF. As for the peak demand from 2038 to 2050, it has been extrapolated from the previous data. The following graph depicts how the demand for electricity is expected to grow over the time period of interest (base case).

¹ Version 6.7.6

² Version 2012

³ Version 2.0

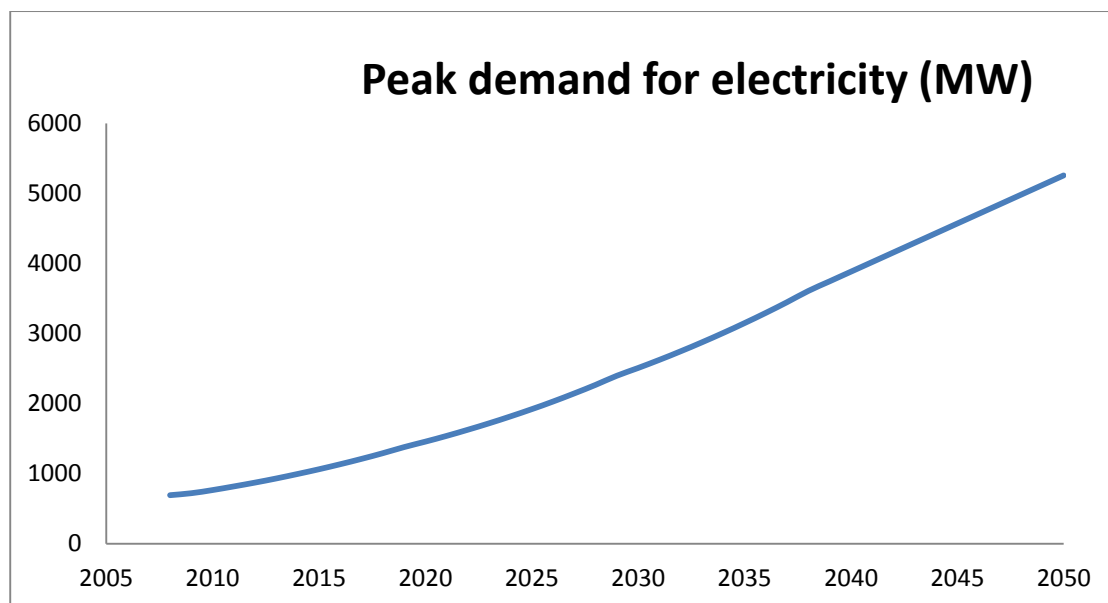


Figure 1. Electricity peak demand growth 2010-2050

As the demand is not homogeneous over periods within a year, the year is split in time slices. Each one of these time slices represents a certain percentage of energy and time.

Time Slice	Percentage of energy and time
Winter Day	12.65
Winter Night	12.65
Long Autumn Day	20.55
Long Autumn Night	20.55
Spring Day	8.35
Spring Night	8.35
Summer day	8.45
Summer Night	8.45

Table 1. Time Slices of the model. This split is based on data from a study on Kenya (due to similarities and lack of information on Tanzania) done at the Division of Energy Systems Analysis at KTH (**KTH-DeSA, 2012**).

2.2 Technology

The tables below show some indicative features of the technologies that are considered in the current model. Currently, the installed capacity in the country is equal to 2003 MW of which 46% is hydro and 34% natural gas (SNC-LAVALIN-INTERNATIONAL-INC, 2011).

Technology	Capital Cost	Variable Cost	Fixed Cost
	\$/kW	\$/kWh	\$/kW
Coal Plant	2000	0.0052	60
Natural Gas Plant (CC)	750	0.0031	36
Diesel Plant	469	0.0099	125
Biomass Plant	1256	0.0040	46
Small Biomass Plant	1256	0.0040	46
Large Hydropower Plant	2000	-	10
Small Hydropower Plant	2000	-	16
Large Wind Farm	1439	-	40
Small Wind Farm	2370	-	40
Concentrated Solar Plant	3785	-	53
PV	4000	-	27
Geothermal Plant	2238	-	35
Imported Electricity	-	0.2	-

Table 2. Costs of new technologies (Chubu-Electric-Power-Company, 2012)

The data that is shown in Table 2 and 3 is taken from the Model for Electricity technology Assessments (META) created by Chubu Electric Power Company. The reliability of this data derives from the fact that this study is part of the Energy Sector Management Assistance Program (ESMAP) coordinated by the World Bank and it provides information on electricity expansion (The-World-Bank-Group, 2012).

Technology	Life Time	Capacity factor	Efficiency
	Years	%	%
Coal Plant	35	80	40
Natural Gas Plant (CC)	25	75	55
Diesel Plant	25	80	35
Biomass Plant	25	75	35
Small Biomass Plant	25	75	35
Large Hydropower Plant	80	60	-
Small Hydropower Plant	25	45	-
Large Wind Farm	25	25	-
Small Wind Farm	25	25	-
Concentrated Solar Plant	25	20	-
PV	25	20	-
Geothermal Plant	25	80	-

Table 3. Technical features of new technologies

In this model, it is assumed that capacity and availability factors in Tanzania do not depend on the time slice apart from the case of solar power technologies since in the night time slices there is obviously no sunlight. By capacity factor of a power plant

we mean the annual produced energy over the energy that it would produce if it ran at its maximum capacity over the entire year period. Availability factor is defined as the share of time when a power plant is available in a year. Time slices are seasonal divisions into which we can categorize different days or seasons of the year.

The transmission and distribution losses are assumed to be constant and equal to 17% (Index-Mundi, 2010).

2.3 Fuel

In the current model the scenarios that are developed are hypothetical as the main objective is the comparison between the energy modeling tools. These scenarios are based on global fuel price projections. It has to be noted that a study focusing on energy policy design should investigate further fuel options based on different databases and take into account the fact that there might be local resources that can be extracted at a lower cost.

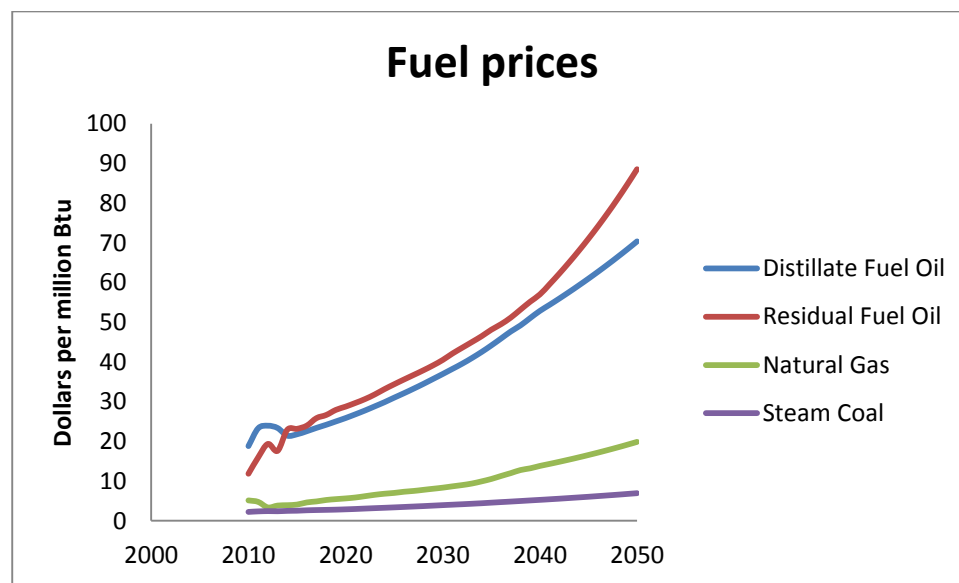


Figure 2. Projection of main fossil fuel prices (EIA, 2013)

The fuel price for renewable sources is assumed to be zero except for biomass. The biomass price is 0.06 \$/KWh (IRENA, 2012) and as it is assumed not to be depleting, this price is fixed all over time.

3. Model description

In this section, a description of how the model is built is given. It should be noted that these inputs help build scenarios, which are hypothetical (but internally consistent)

views of the future. As some of this data is based on the author's best guess, it is also subject to a detailed sensitivity analysis in Appendix B4.

- The time period of the simulation starts from 2010 and ends in 2050 with constant 1-year step.
- The time-slices were defined based on a previous study for Kenya (KTH-DeSA, 2012) due to lack of data for Tanzania. The electricity imports and the efficiency of the transmission and distribution system are constant over the entire period based on 2010 data.
- There are 34 existing and future, individual power plants which are phased out according to their standard lifetime. For example, if a natural gas power plant is built in 2008 (considering a standard lifetime of 25 years) it will be phased out in 2033.
- All the individual power plants have a lower limit on their annual production which is equal to their firm production multiplied by the corresponding coefficient for each particular scenario. These coefficients can be found in Table A.8.
- The new technologies that are considered in the model are divided into two categories. The first category (centralized energy system) includes: natural gas fired plants, coal fired plants, biomass fired power plants, large hydro, large wind farms, CSP and geothermal power plants. The second one (decentralized power plants) includes small biomass fired plants, diesel fired power plants, small hydro, small wind farms and PV. This is consistent with the SAPP study done by IRENA (Miketa, 2013).
- New investments are limited to 0 GW until 2015 and to 1 GW per technology per year for the centralized technology and to 0.5 GW for the decentralized ones from 2016 onwards. By doing this it is attempted to avoid huge investments in a one year and or one technology which would be not realistic.
- The capital cost of the transmission and distribution system is considered to be 10% of the respective cost of hydro. This is consistent with assumptions made in another study on electricity expansion in Africa (Taliotis, n.d.).
- The decentralized power plants are assumed to bear no transmission costs or losses. (In the model they are represented by being connected to a second transmission and distribution system with zero cost and 100% efficiency. This is to simulate no transmission losses as these plants are assumed to be located close to demand and it also allows easier modelling.
- The discount rate chosen in the model is 10%, which is a value that has been used in other relative studies (Miketa, 2013).

The current model of the electricity sector of Tanzania is replicated in 3 pieces of software. Although the main modeling principles are common, some features have to get adapted to the particular case of LEAP:

- In LEAP it is not possible to have a lower production limit and allow optimization in the same technology simultaneously. Thus, two modules⁴ are required for the individual power plants; one for the forced production and one for the simulation. Each technology has to be split into two parts with equal capacity but with different capacity factors. An example is given:

Assume a technology with a capacity of 1 GW, a capacity factor of 0.8 and a lower annual production of 3000 GWh. All these pieces of data are entered in the module that represents the lower production. In the module the simulation module, the capacity factor is calculated as following:

$$cf=(1*0.8*8760-3000)/(1*8760)=0.46$$

- The dummy TnD technologies which link the electricity with the final electricity demand and the electricity exports respectively (shown in Figure 3) are not necessary in LEAP.
- In the output properties tab in the output fuels, the “Import fuel to meet shortfall” option has to be selected in the case of electricity and decentralized electricity. Moreover, the cost of these fuels has to be extremely high (999999 USD/Gigajoule) so to make direct fuel imports not a desirable option.

Finally, in MESSAGE it is not possible to set the construction time of a technology equal to zero while in the other two pieces of software it is not possible to set a construction time at all. As the construction time affects the capital cost of a technology rendering the net present value, this issue was offset as following. In MESSAGE the construction time was set equal to 1 year while in LEAP and Answer-OSeMOSYS the capital cost of the new power plants and the TnD system was discounted for a one year time period using the following formula:

$$NPV = \frac{1}{Building\ time} * Overnight\ Cost * \sum_{Years} \frac{1}{(1 + DR)^{Yr}}$$

⁴ A module is a technology or a set of technologies that produce the same fuel at the same stage, i.e. power plants that produce electricity that feeds the TnD system.

3.1 Reference Energy System

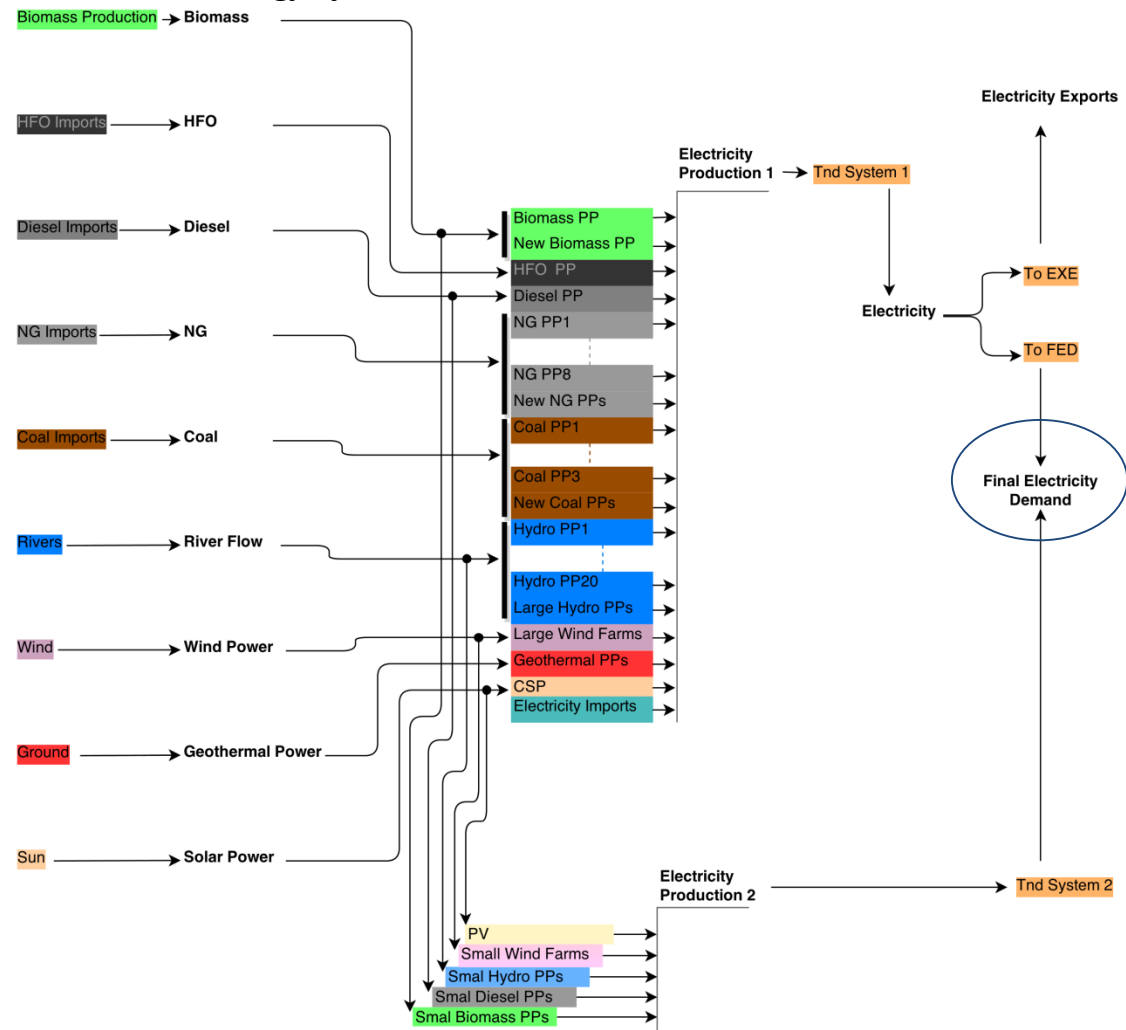


Figure 3. Depiction of the Reference Energy System of the model

3.2 Scenarios

Below the three different scenarios (production wise)⁵ are described. For each of these cases, three sub-cases based on different energy demand projections are considered.

3.2.1 Scenario 1 – Centralized System

This scenario takes into account the existing and potential future power plants considering their capacity and their firm production. In order to cover the gap between the production and the demand, new investments take place in large technologies which are connected to the central transmission and distribution system. Moreover, some exports which are assumed to be 10% of the demand are considered from 2016

⁵ By “production wise” we mean that there is no difference on the final demand side but the production pattern changes for the three different cases.

onwards. Subject to the new build constraints the model is free to choose the least cost investment options.

3.2.2 Scenario 2 – Decentralized Power Plants

In this case, the existing and the future individual stations operate as they do in Scenario 1, but new investments are attempted to take place in small, decentralized power plants (which are not considered in Scenario 1) with no transmission cost and losses, since these are supposed to produce electricity that is used directly. The penetration of decentralized power generation in the system is gradual and it is achieved by setting a lower production limit in the second TnD system (see Appendix A for the corresponding limit).

3.2.3 Scenario 3 – 100% renewable

The goal of this scenario is to promote investments solely in renewable energy technologies. All the existing and the future fossil fuel fired plants are phased out 5 years earlier compared to the previous cases as well as the two coal fired plants (see the Reference Energy System in Figure 3) that are planned to be built in 2024 and 2025 are not taken into account. The new technologies that are considered are both centralized and decentralized but all the fossil fuel based technologies are excluded. The contribution of centralized and decentralized technologies to the final electricity demand is attempted to be almost equal by the end of the period. The exports are similar to the first case.

4. Results and discussion

In this section, graphs that depict the share of each energy source in the final electricity generation are deployed. These graphs correspond to the base demand projection for the three (production based) scenarios, while for the low and the high projection, the results are shown in Appendix B.

It should be mentioned that (at this point in its development) LEAP has two limitations that have an impact on the modeling process. Firstly, historical capacity and optimization option cannot be selected in the same module. Thus, a new module was created to facilitate the modeling of individual power plants. Apart from that, LEAP is capable of optimizing only one module. It must also be mentioned that LEAP's optimization capacity is continually developing, so this issue is likely to be overcome shortly.

4.1 Scenario 1

Comparing the results that derive from each modeling case it is observed that these are almost identical. Hydro seems to be the dominant technology since its potential in Tanzania is huge. Answer-OSeMOSYS shows a small penetration of wind (around 11% in 2050) in the end of the modeling period while LEAP makes some investments in geothermal at that time that don't exceed 7%. This difference can be explained most probably by the fact that the model in LEAP is set up in a slightly different way due to the afore mentioned limitations.

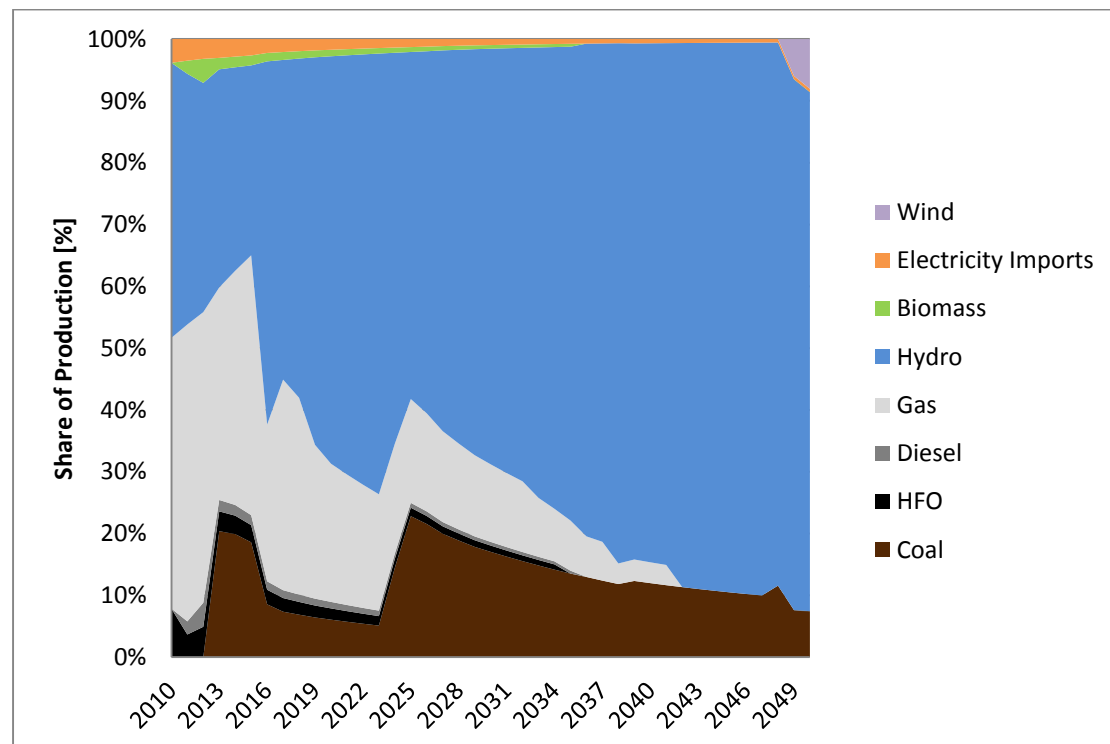


Figure 4. Answer-OSeMOSYS, Scenario 1

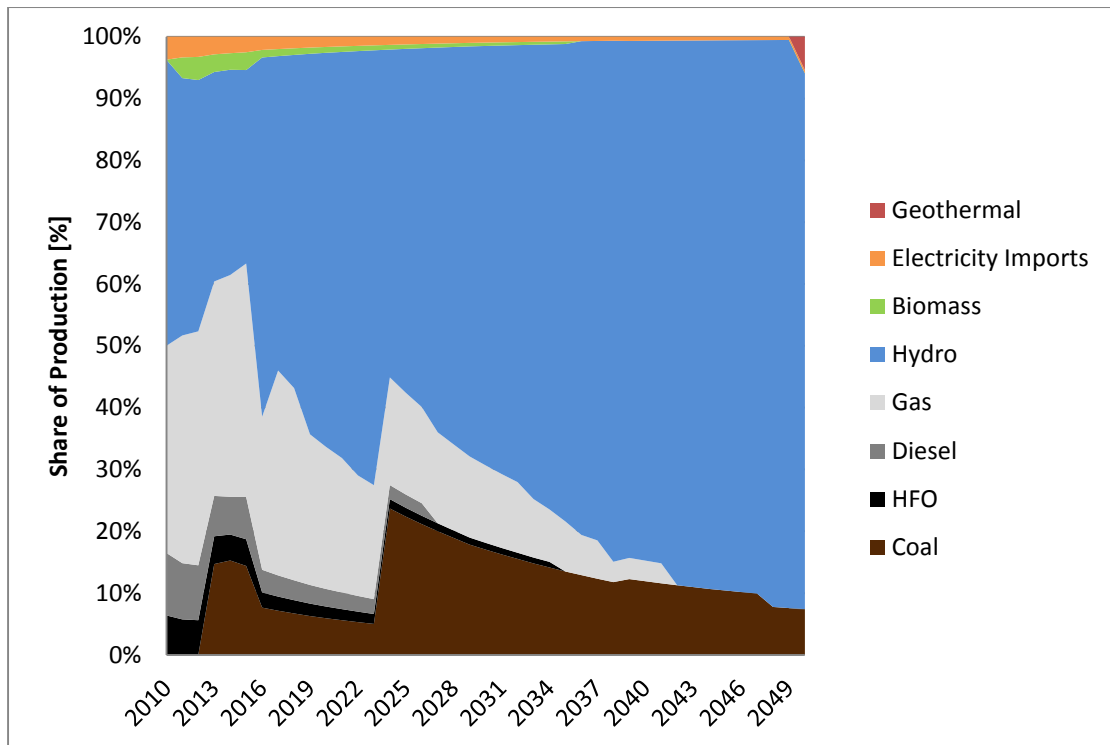


Figure 5. LEAP, Scenario 1

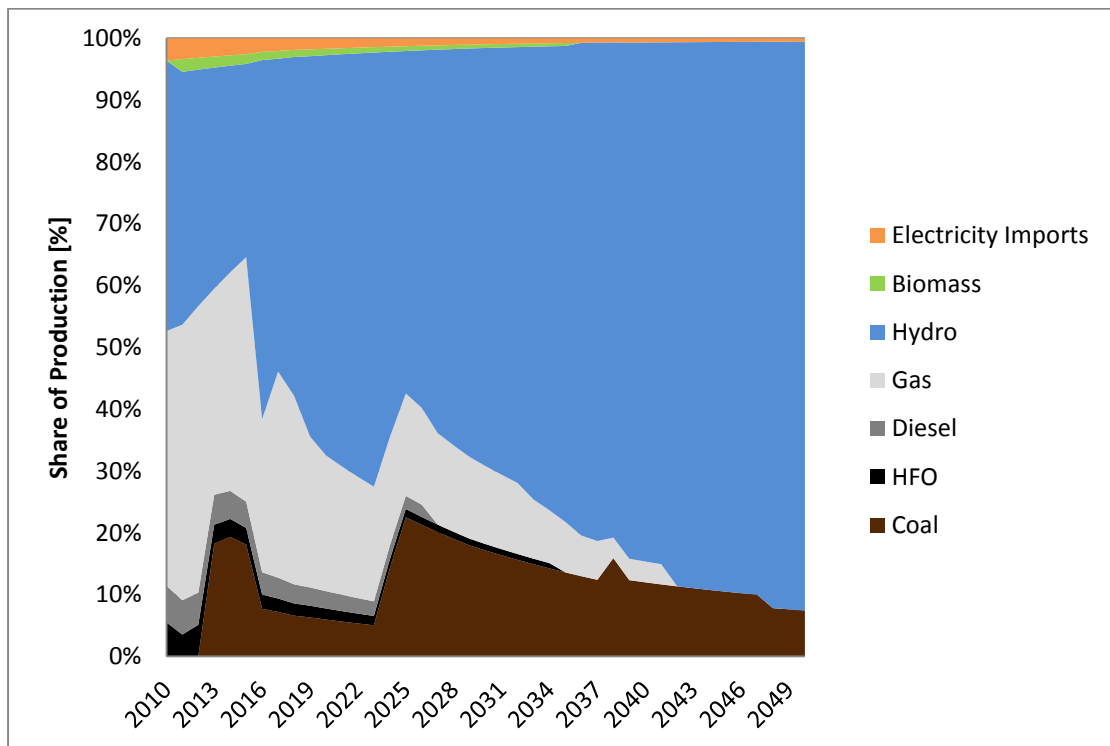


Figure 6. MESSAGE, Scenario 1

4.2 Scenario 2

In this case, the results from the different tools used in this study are also quite close for the centralized system. Regarding the decentralized technologies, all the pieces of software invest in hydro but MESSAGE has a small share of wind in the latest years of the modeling period.

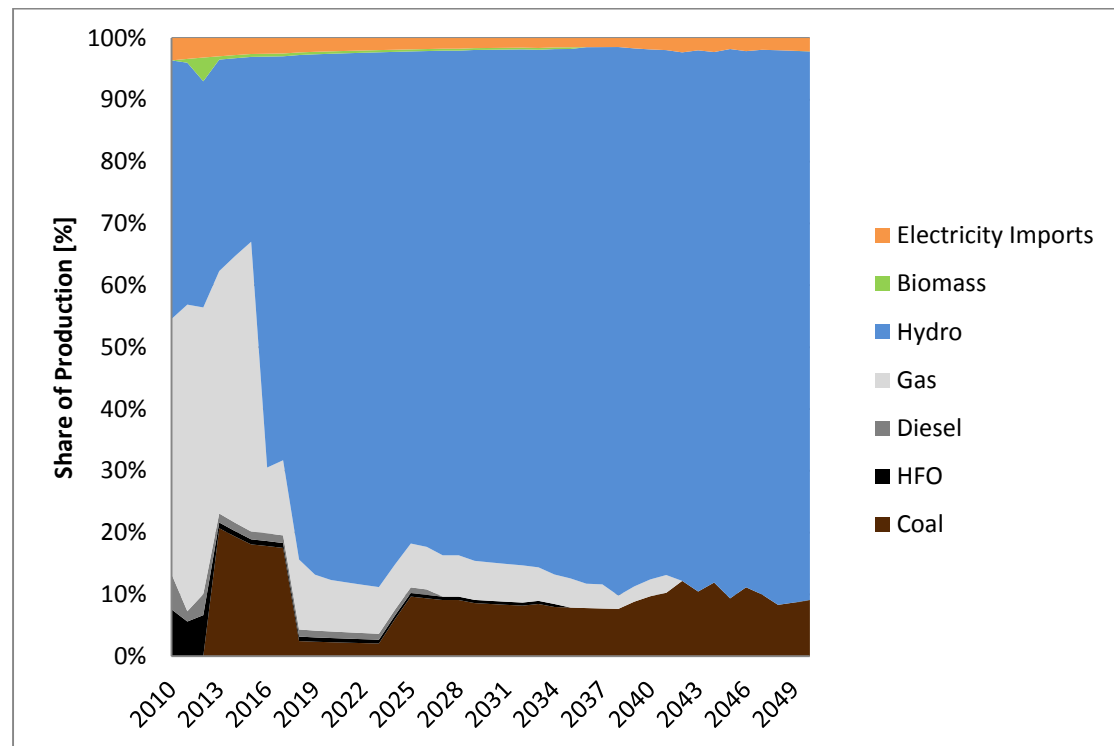


Figure 7. Answer-OSeMOSYS, Scenario 2, Centralized system

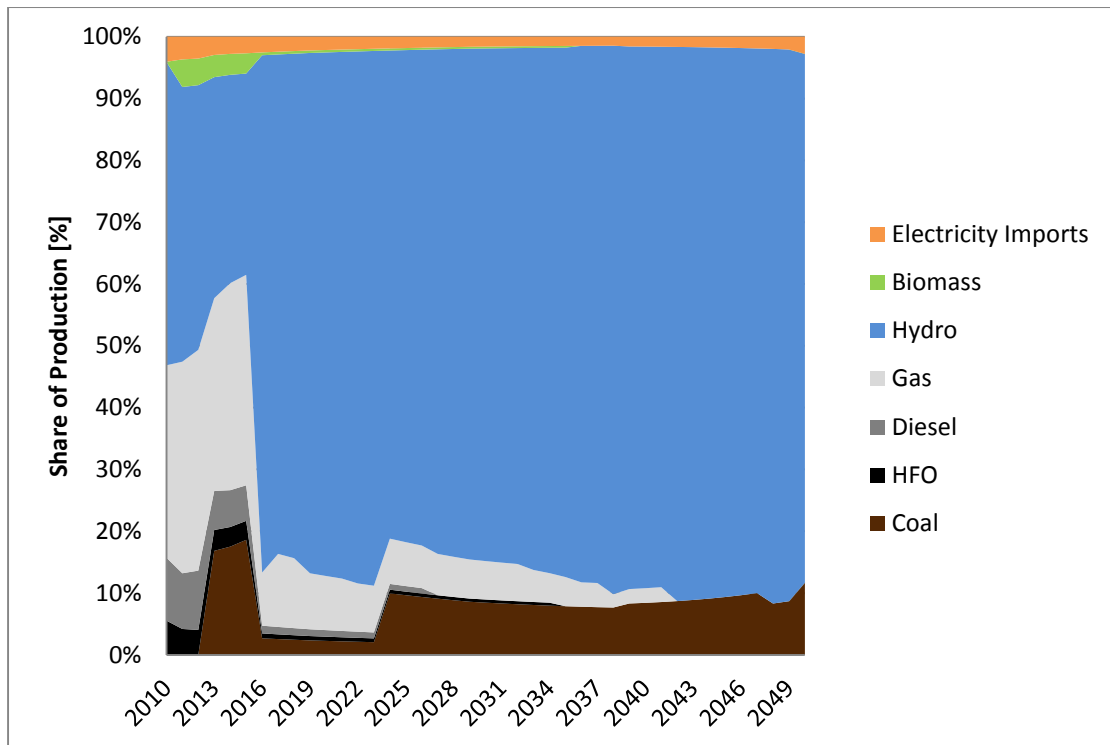


Figure 8. LEAP, Scenario 2, Centralized system

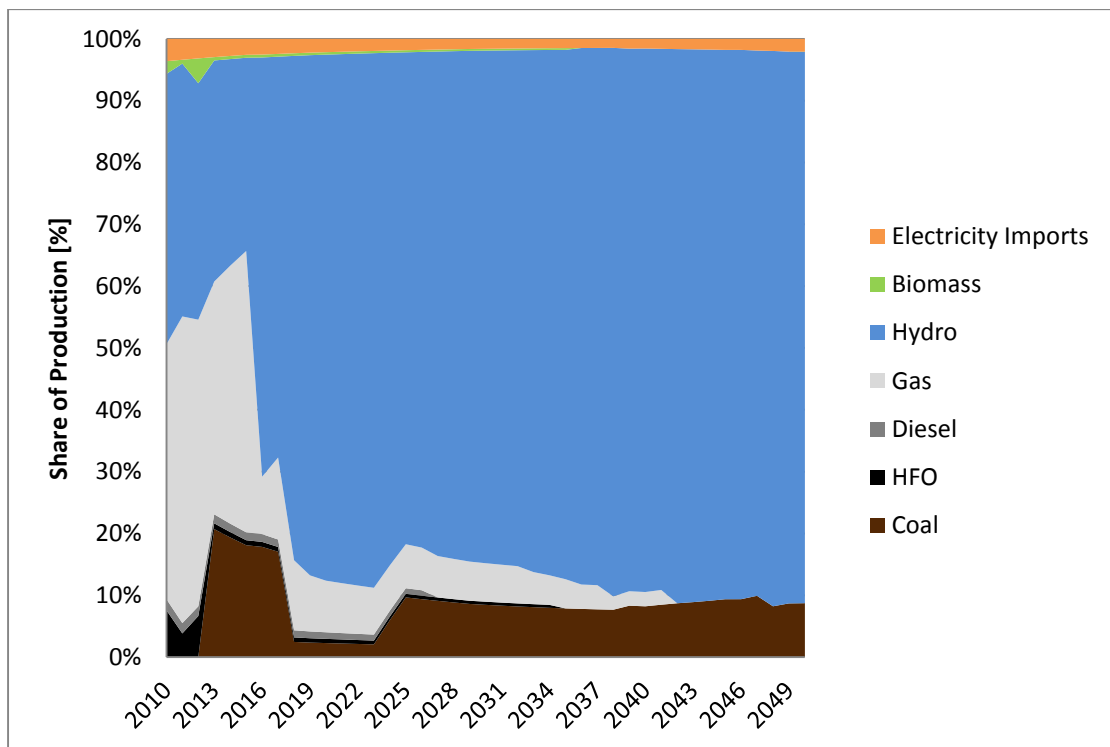


Figure 9. MESSAGE, Scenario 2, Centralized system

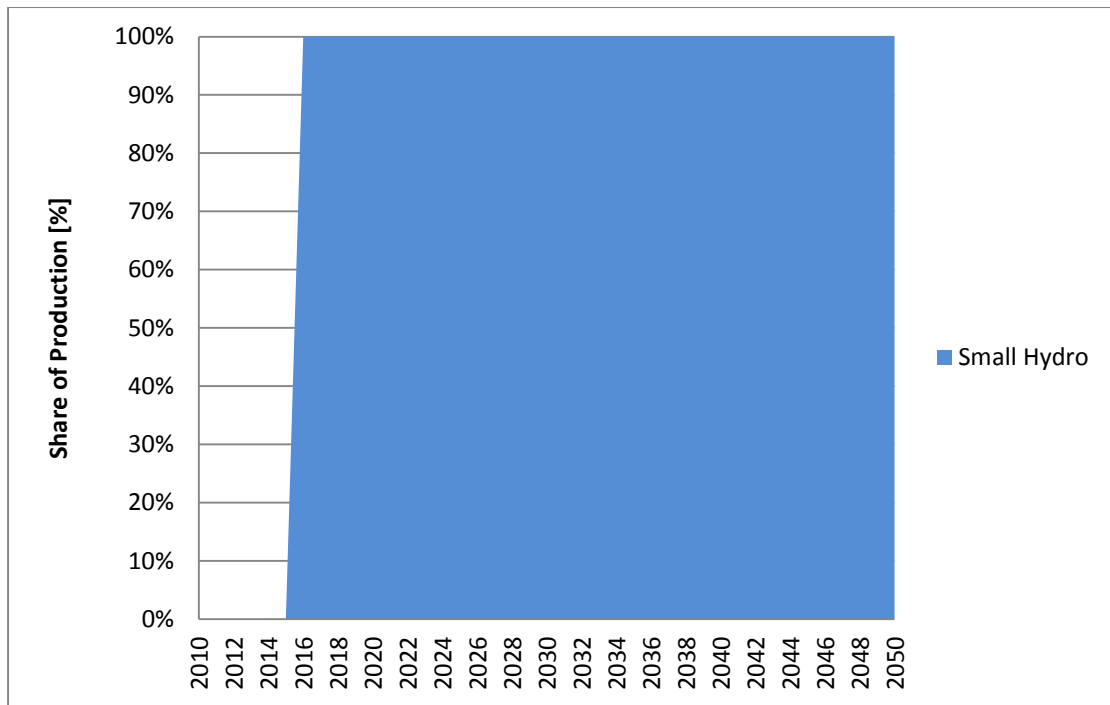


Figure 10. Answer-OSeMOSYS, Scenario 2, Decentralized technologies

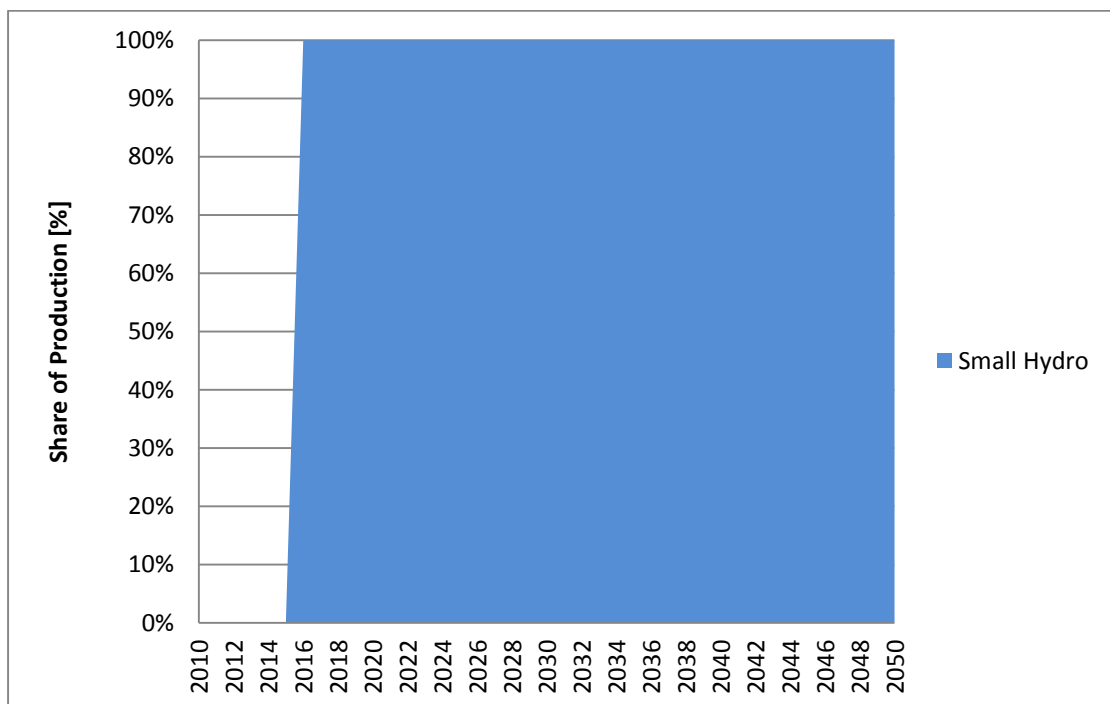


Figure 11. LEAP, Scenario 2, Decentralized technologies

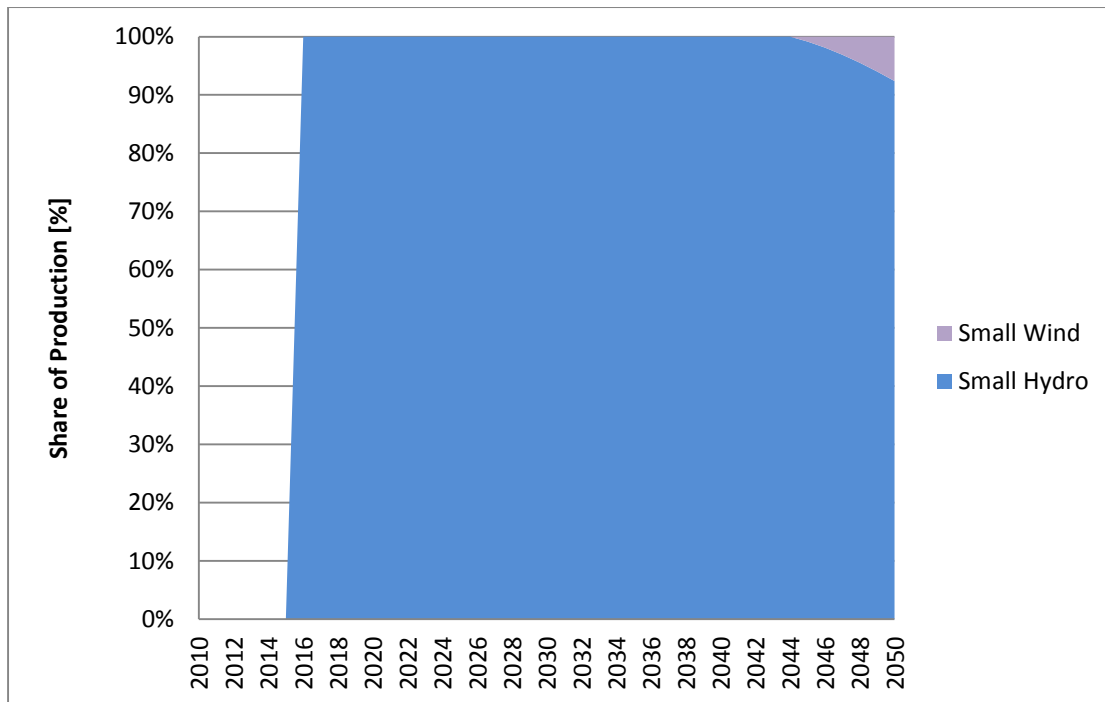


Figure 12. MESSAGE, Scenario 2, Decentralized technologies

4.3 Scenario 3

In the case of the renewable scenario, the results for the centralized system are again very close (slight differences appear in small periods of time) while for the decentralized system are 100% similar. More specifically, the share of biomass in LEAP from 2012 to 2017 fluctuates between 2% and 5% while in the other models it is constant at 1%. Furthermore, the share of coal in LEAP from 2012 to 2014 is 4-6% lower compared to the other cases as well as natural gas from 2032 to 2037 contributes 7% to the power generation compared to 2% in MESSAGE and Answer-OSeMOSYS. Finally, the most distinctive difference between MESSAGE and Answer-OSeMOSYS is a spike in the coal curve in the year of 2038 which shows an increase of 5% compared to the previous and the next year.

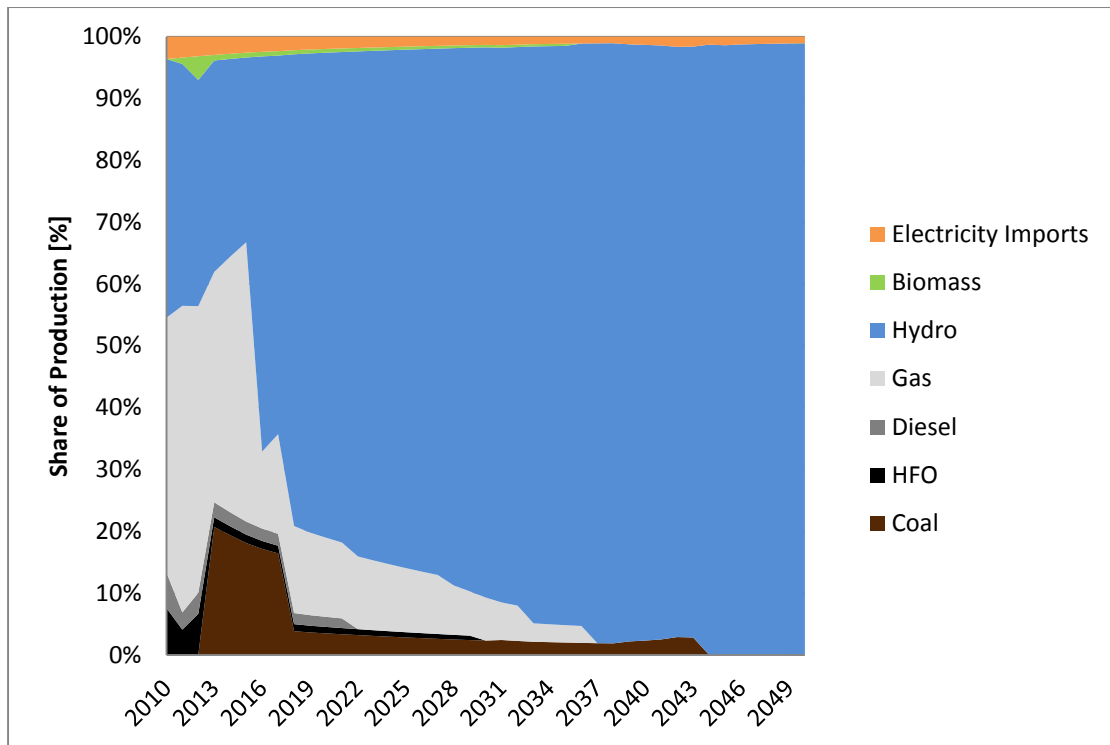


Figure 13. Answer-OSeMOSYS, Scenario 3, Centralized system

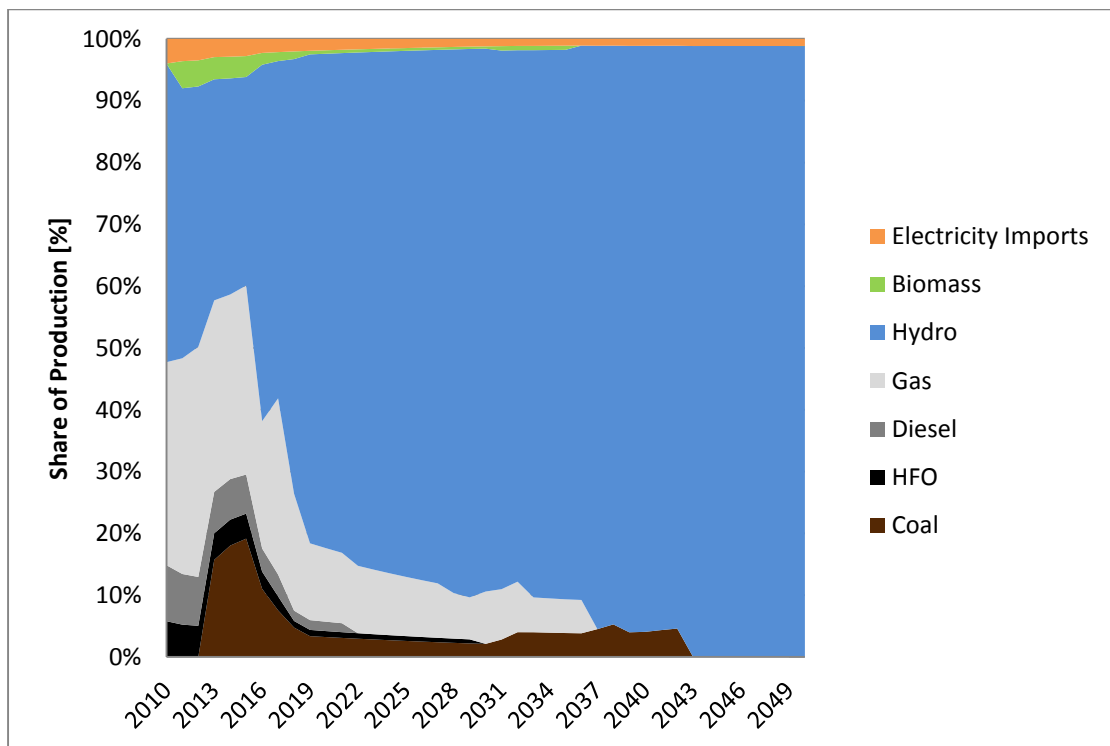


Figure 14. LEAP, Scenario 3, Centralized system

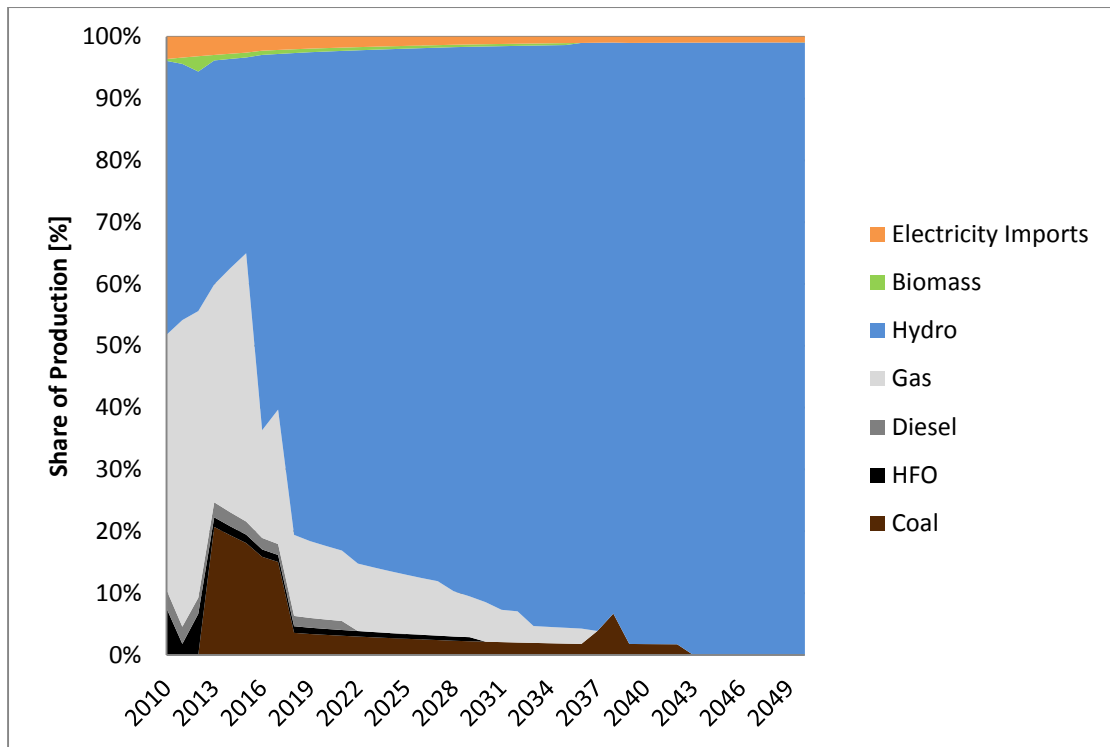


Figure 15. MESSAGE, Scenario 3, Centralized system

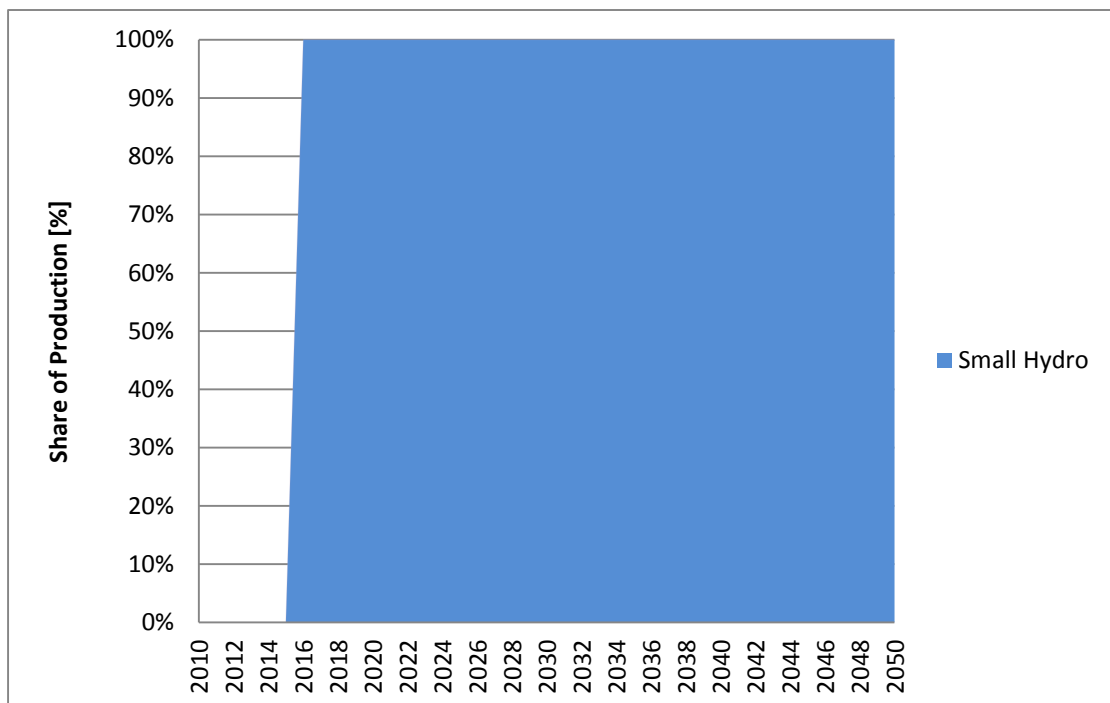


Figure 16. Answer-OSeMOSYS, Scenario 3, Decentralized technologies

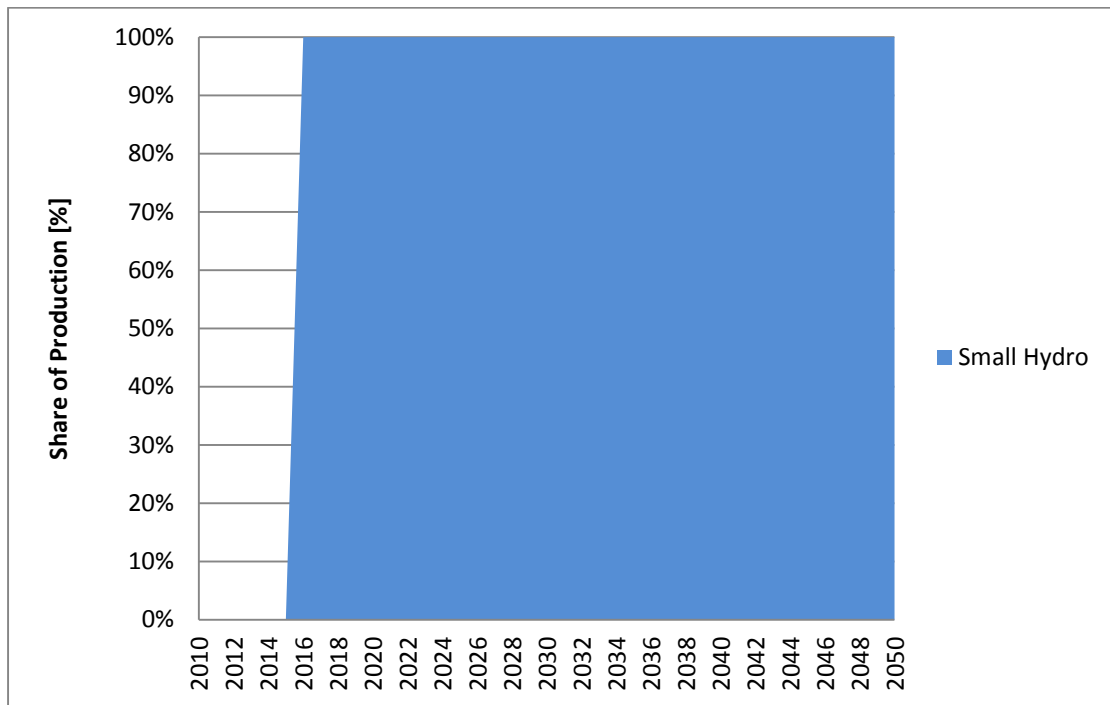


Figure 17. LEAP, Scenario 3, Decentralized technologies

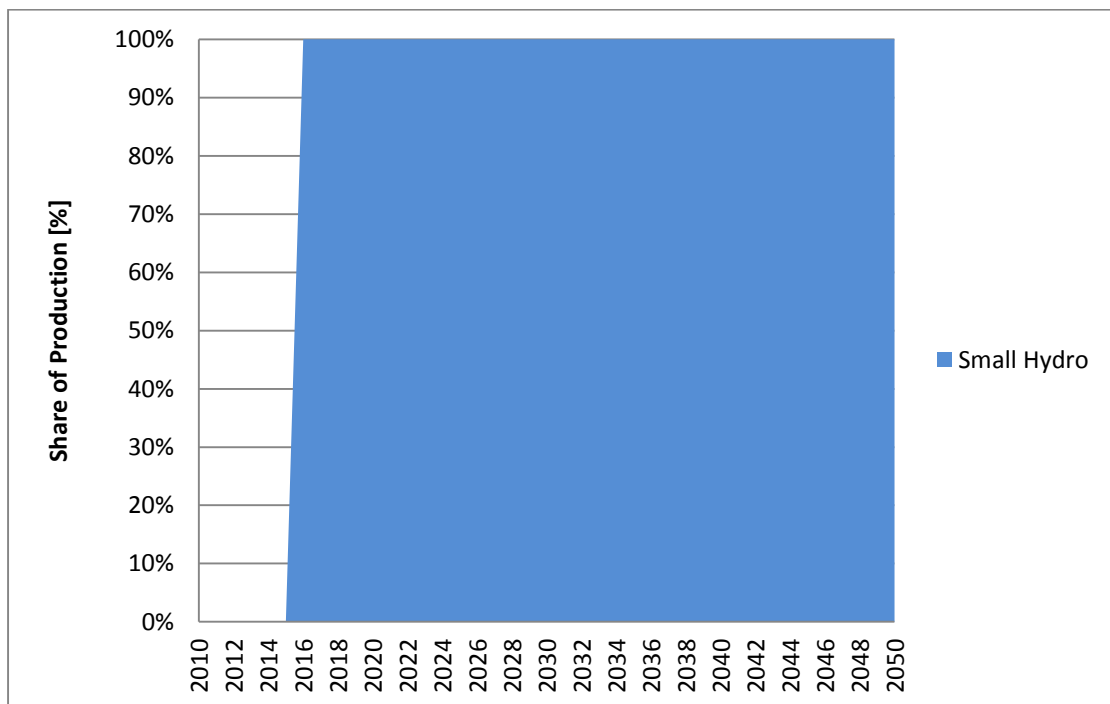


Figure 18. MESSAGE, Scenario 3, Decentralized technologies

5. Funding opportunities and geopolitical aspects

5.1 Funding opportunities

As mentioned in the introduction, finding funding for the projects aiming to help the low carbon electrification transition in Tanzania is very important. These kinds of projects are numerous all over Africa. In this section, some hints are given to find such funding.

5.1.1 Public funding

Often, wealthy countries have a Development Agency or specific programs which aim at providing funding to help developing countries. The Australian Government Aid Program, Danish International Development Agency (**DANIDA**), the UK Department for International Development (**DFID**), the International Fund for Agricultural Development (**IFAD**), the Netherlands Ministry of Foreign Affairs (**NMFA**), and the Swedish International Development Cooperation Agency (**Sida**) are examples of such Development Agencies. In this paragraph, two Development Agencies which participate actively in African project funding are discussed: the Abu Dhabi Fund for Development and the French Development Agency. These two particular cases are chosen because they have a high turnover (see below), they are significantly active in Sub-Saharan Africa (Abu-Dhabi-Fund-for-Development, 2013) (Agence-Française-de-Développement, 2013) and they often carry out a cost-benefit analysis before funding a project (such an analysis is done the current study).

- **the Abu Dhabi Fund for Development**

Abu Dhabi Fund for Development (ADFD) was established as a leading national entity to provide development aid to developing countries in the form of concessionary loans and administering grants on behalf of the Abu Dhabi government. The aim is to finance development projects in order to support efforts to achieve economic and social development in developing countries. ADFD also makes investments in order to encourage the private sector in the recipient countries to play an essential role in accelerating the economic development process.

The total value of loans, grants and investments funded by ADFD and the Government of Abu Dhabi during the past four decades, reach almost AED 35 billion for 325 development projects in 59 countries around the world. (Abu-Dhabi-Fund-for-Development)

- **French Development Agency (AFD)**

The AFD is a public institution providing development financing. A specialized financial institution, AFD finances sustainable development projects carried out by government local authorities, public companies, and the private and associative sectors in five continents—with primacy given to Africa which receives two-thirds of AFD commitments—and in overseas France. These projects focus on urban development and infrastructures, rural development, industry, financial systems, education and health. AFD offers a range of financial instruments: subsidies,

guarantees, shareholdings, and all forms of assisted and commercial loans—in response to each situation (Agence-Française-de-Développement).

5.1.2 Private funding

- **The Sustainable Energy Fund for Africa (SEFA)**

SEFA aims to support the implementation of the African Development Bank's Private Sector Development (PSD) Strategy by providing untied grants and equity support to smaller-size renewable energy and energy efficiency players. It is also envisioned that SEFA will support the Bank's energy strategy, which is currently under development. It aims at exploiting the potential of the emerging "clean energy" market in Africa as a source of growth and innovation for SMEs as producers, distributors, suppliers, and consumers of climate-friendly energy. In the greater context of the Africa Commission, this initiative and associated fund are seen as a means to achieve higher employment and induce competitive job creation throughout Africa. (African-Development-Bank-group, 2013)

SEFA will provide two financing windows. The first will support the development and design of renewable energy generation and distribution projects with a total investment between USD 30 and 75 million on a grant basis. The second will support SMEs all along the renewable energy and energy efficiency value chain with a total investment of under USD 30 million on a mixed grant and equity investment basis.

- **The Climate Technology Initiative Private Financing Advisory Network (CTI PFAN)**

The Private Financing Advisory Network (PFAN) is a multilateral, public-private partnership initiated by the Climate Technology Initiative (CTI) in cooperation with the UNFCCC Expert Group on Technology Transfer. PFAN operates to bridge the gap between investments and clean energy businesses. (CTI-PFAN, 2013)

- **The Africa Enterprise Challenge Fund (AECF)**

AECF is a US\$190M private sector fund, backed by some of the biggest names in development finance and hosted by the Alliance for a Green Revolution in Africa (AGRA). Its aim is to encourage private sector companies to compete for investment support for their new and innovative business ideas. (Funding-Innovatio-for-Business-in-Africa)

- **The Climate Investment Funds (CIF)**

The CIF are comprised of two Trust Funds, each with a specific scope and objective and its own governance structure. (Climate-investment-fund)

- o The Clean Technology Fund promotes investments to initiate a shift towards clean technologies. The CTF seeks to fill a gap in the

international architecture for development finance available at more concessional rates than standard terms used by the Multilateral Development Banks (MDBs) and at a scale necessary to help provide incentives to developing countries to integrate nationally appropriate mitigation actions into sustainable development plans and investment decisions. Through the CTF, countries, the MDBs and other partners agree upon country investment plans for programs that contribute to the demonstration, deployment and transfer of low carbon technologies with significant potential for greenhouse gas emissions savings.

- o The Strategic Climate Fund serves as an overarching fund to support targeted programs with dedicated funding to pilot new approaches with potential for scaled-up, transformational action aimed at a specific climate change challenge or sectorial response.

5.1.3 Clean Development Mechanism

The Clean Development Mechanism (CDM) is a flexibility mechanism defined in the Kyoto Protocol. This mechanism is based on the trading of Certified Emission Reduction units. Certified Emission Reduction units are a type of emission unit or carbon credits (Betz, 2007).

The CDM allows the industrialized countries to meet part of their emission reduction commitments under the Kyoto Protocol by buying Certified Emission Reduction units from CDM emission reduction projects in developing countries. It is better for industrialized countries to lead CER projects in developing countries because it is less expensive to carry out than in industrialized areas. (FAO)

In short, the CDM intends to meet two objectives through these emission reduction projects in developing countries financed by industrialized countries:

- to assist developing countries in achieving sustainable development and in contributing to the ultimate objective of the United Nations Framework Convention on Climate Change which is to prevent dangerous climate change;
- to assist industrialized countries in achieving compliance with their quantified emission limitation and reduction commitments (greenhouse gas emission caps).

Between 2001, which was the first year when CDM projects could be registered and September 2012, the CDM issued 1 billion Certified Emission Reduction units (CDM, 2012). The new CDM projects are nowadays predominantly renewables and energy efficiency projects.

5.2 Geopolitical key facts related to hydrocarbons

Large scale energy planning involves aspects such as social conditions or geopolitical status that cannot be quantified. However, they must be taken into account since they can affect significantly the final decision. Geopolitical dynamics can have an impact on the cost of energy production, on the source availability as well as on the technology that can be utilized (i.e. nuclear non-proliferation). Tanzania is an East African country with a colonialist past. It is considered to be a geopolitically sensitive area and every decision has to be interrogated very carefully.

Monitoring the case of Uganda where recently a significant number of barrels of exploitable oil was discovered, Tanzania prospects to launch its own exploration and exploitation plan (via foreign investments). In some parts of the country, all the territorial border issues with the neighboring countries have been solved successfully. Nonetheless, in Lake Malawi/Nyasa -a location of high interest in terms of hydrocarbon exploitation- there is a long history, unresolved dispute with Malawi. The latest claims rights over the entire lake while the Tanzanian side asserts that 50% of the lake waters should be under its sovereignty. Malawi granted licenses for hydrocarbon exploration in the eastern part of the lake in September 2011. This action lead to the escalation of the conflict between the two countries. On both sides there are actors who claim that a peaceful solution, beneficial for both countries could be achieved at the same time when far right forces approach the issue from a more aggressive point of view. The resolution of the dispute could play a catalytic role in the design of energy policy of Tanzania over the next few years. (Open-Democracy, 2012)

The case of natural gas reserve discoveries and exploitation is on better track compared to the case of oil. Tanzania has recently discovered some 934 cubic meters of offshore natural gas. Very big companies such as Statoil, Total, BP and Shell prospect to come to agreement with the government of Tanzania laying the foundations for making the country an attractive destination for large foreign investments. A share of the country's natural gas reserves is consumed domestically in the electricity sector as well as in industrial activities (CS-monitor, 2013). The opportunities for NG exports are great since the demand is growing especially in the emerging markets of Asia (mainly in China). Due to the fact that Tanzania has not proximity to major natural gas importers, the most prominent technological solution to facilitate the exports is the construction of LNG stations (Euractiv, 2013).

5.2.1 Implications

Based on the above mentioned facts, five main impacts of the hydrocarbon exploitation on the Tanzanian economy and politics can be noted. Firstly, the access to low cost, domestic fossil fuels could make renewables uncompetitive and consequently the energy policy could utterly change. Additionally, significant amounts of hard currency could be obtained through the exports which could be used

to stimulate national economy. Apart from that, infrastructure projects such as roads and ports that would be potentially constructed to facilitate the trade of hydrocarbons could be exploited properly by the state to boost economic growth. Moreover, a successful hydrocarbon trade agreement could establish further trade relations with other countries making Tanzania a robust market economy. Finally, the use of natural gas for cooking substituting charcoal could lead to a reduction (and potentially to the elimination) of the deforestation rate.

5.2.2 Political stability and the effects of mining activity

Tanzania has been an independent state since more than 50 years ago with multi public democracy for about 20 years without experiencing any military coup or civil war (Epza, Tanzania political stability, 2010). Political stability is a very important factor for a country to attract foreign investments and exploit its natural resources. Hence, the current political status of the country seems to be ideal. Tanzania has also a rapidly growing mining sector (Epza, Market access vision, 2010). This type of industrial activity requires large investments in infrastructure especially in the energy sector. Therefore, power plants that could be built by foreign investors for empowering the mining sector could be integrated to the national energy system.

5.2.3 Regional integration and water policy

Based on the least cost optimization modelling it turns out that the dominant energy source in Tanzania is hydropower. However, there are factors that could inhibit huge investments in hydropower plants. In Africa, a large discussion on regional infrastructure integration has opened and more specifically Tanzania could potentially be a part of the Eastern Africa Power Pool (PIDA, 2011). This implies that decision making requires consensus and it cannot be made by one country itself. Thus, different approaches in policy might arise between the country members that could utterly change the energy strategy. Apart from that, large investments in hydro could be also limited due to the fact that they require high capital cost at the same time that the weather patterns are quite uncertain due to climate change. Finally, as the water resources are needed to cover national needs, an approach from a purely energy point of view is not enough to give insight into water management strategies (Welsch, 2014). Taking these facts into account, investments in natural gas could be a more foreseeable option.

6. Conclusions

Taking a close look at the results it can be observed that hydropower is the dominant energy source in all the scenarios. In order to assess the strengths and weaknesses of each scenario, there are factors that must be taken into account that cannot be

modeled, such as the transfer of know-how. It is quite important to investigate how could the local communities participate actively in the decision making process in the case of decentralized technologies. Undoubtedly, the management of the power plants would be easier on a national level (centralized system). From a technical point of view, the main difficulty in the case of the centralized system is the construction of a robust power grid. On the other hand, the off grid technologies (most often renewables) face the problem of intermittency. Thus, storage options are required and must be investigated. This has an impact on the overall investment cost.

This study has examined different patterns for the electricity sector in Tanzania. However, in order to design an entire energy system properly a holistic approach is required. This approach should take into consideration other energy sub-sectors such as heat demand and transport and the interactions between those. Furthermore, the investigation and modeling of the interlinkages between energy sector and land or water use (as well as other related systems) could give even more beneficial results for a strategic development planning. Additionally, from an economic point of view the cost of energy does not form a sufficient indicator for the determination of the contribution of an energy system to a national (or local) economy. Other factors like job creation or impact on infrastructure development must be considered. A computable general equilibrium model (CGE) could facilitate the design of an energy system considering those aspects. Furthermore, energy planning must take into account geopolitical aspects as those can have an influence on several parameters (explained above). In the current study, the energy system design is based on given demand projections. However, it must be noted that in case there is funding for electricity generation plants (such opportunities have been investigated), the demand could be re-calculated based on the production. More specifically, instead of defining the electricity demand and considering this fixed building power generation technologies, a given generation potential could define the demand.

Regarding the tools that have been used in this study, all these generated very close results. It is the opinion of the author (and based on the limited experience gained in this study) that each case revealed strengths and weaknesses. The main advantage of OSeMOSYS is the fact that it is an open source and it is governed by transparency and expandability. The next comments apply to user interface issues. ANSWER-OSeMOSYS is a user friendly tool since the interface is compatible with excel in a way that cells can be easily copied, pasted and subsequently treated. Its main weakness is the low level of flexibility with units. Relationships between units must be defined by the user. This issue should be properly addressed in a future version. On the other hand, LEAP is very versatile with units but importing data from excel (while possible and functional) is a little more complicated. One of its great advantages is its wide fuel database. What has to be improved is mainly with respect to the optimization part. More specifically, in LEAP it is not possible to enter historical production in a technology in a module that is optimized. Apart from that LEAP is capable of optimizing only one module per run. Moreover, it is not possible to enter a

capital cost and a lifetime for transmission and distribution system. Thus, this has to be integrated to the energy producing technologies. However, since the lifetime of the power grid in most cases is not exactly equal to the power plants' the results can be slightly affected. Finally, in terms of user operation, MESSAGE was the fastest piece of software in the optimization process but it was (the limited) experience of the author, that main disadvantage is the fact that it becomes unstable when using some specific sets of units. For instance, when "percentage" is selected as the unit for plant factor it is considered as "share". While this will likely be resolved with the continuous development of the software, the user would be advised to stick with MESSAGE's default units. Its main strength compared to the other two tools (at this point) is its ability of controlling more variables in an advanced and complex way by setting constraints.

Taking everything into account, two main conclusions can be drawn from this study. Firstly, the prospective user should be aware of the limitations of each tool. The ease of use in several cases depends on subjective criteria. The access to learning resources is also a catalytic factor. Regarding energy modeling in general, it is quite important for the user to realize that modelling it is a very useful tool but definitely not enough to make energy policy. Geopolitical and social factors as well as funding opportunities can have an impact on the decision making process. Also, different stakeholders approach the design of energy policy and strategy from different points of view.

7. Suggested future work

Based on the preceding work, selected suggestions are now made. These suggestions intend to either improve the current study or investigate new, relative aspects.

- The model could be re-constructed improving some of the inputs. More specifically, actual climate data could be used to define the capacity factors of the renewable energy technologies.
- Based on the model of the electricity sector, the study could be extended covering the entire energy system of the country.
- A broader study including neighboring countries could be carried out, prospecting to make a strategic policy for a group of African countries.
- The electricity demand could be further analyzed including sub-sectors like households, services and industry.
- The social implications of investing in different technologies could be interrogated. For example, the erection of a large dam could displace a number people.

- A CGE model could be created to evaluate the impact of the investments in different technologies on job creation and other sectors of the economy.
- A CLEW model could be built to examine the interactions between climate, land, energy and water and find a general, optimal solution from a strategic point of view.
- Different storage technologies could be considered and compared such as batteries, hydro pumped storage and hydrogen production.

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Appendices

Appendix A – Modeling inputs

Time Slice	Share of energy and time
Winter Day	0.1265
Winter Night	0.1265
Long Autumn Day	0.2055
Long Autumn Night	0.2055
Spring Day	0.0835
Spring Night	0.0835
Summer day	0.0845
Summer Night	0.0845

Table 0.1. Time slices of a year in Tanzania

Primary	Secondary	Tertiary	Final
Biomass	Electricity Production 1 (EP1)	Electricity	Final Electricity Demand (FED)
Heavy Fuel Oil (HFO)	Electricity Production 2 (EP2)		Electricity Exports (EXE)
Diesel			
Natural Gas (NG)			
Coal			
River Flow			
Wind Power			
Geothermal Power			
Solar Power			

Table 0.2. Fuels considered in the model

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Final Electricity Demand (Base) [PJ/a]	16.5384	17.6904	18.9216	20.2356	21.6468	23.1516	24.732	26.4204	28.2276	30.1572	31.8996
Final Electricity	16.1496	17.0712	18.0468	19.0764	20.1672	21.24	22.4496	23.7276	25.0812	26.5104	27.7452

Demand (Low) [PJ/a]												
Final Electricity Demand (High) [PJ/a]	16.8696	18.2196	19.6776	21.2544	22.9572	24.7932	26.748	28.8576	31.1328	33.588	35.8884	
Exports (Base) [PJ/a]	0	0	0	0	0	0	2.4732	2.642	2.8228	3.0157	3.19	
Exports (Low) [PJ/a]	0	0	0	0	0	0	2.245	2.3728	2.5081	2.651	2.7745	
Exports (High) [PJ/a]	0	0	0	0	0	0	2.6748	2.8858	3.1133	3.3588	3.5888	
Decentralized Electricity Production S2 (Base) [PJ/a]	0	0	0	0	0	0	1.2366	1.8493	2.5405	3.3174	4.1468	
Decentralized Electricity Production S2 (Low) [PJ/a]	0	0	0	0	0	0	1.1225	1.661	2.2572	2.916	3.6068	
Decentralized Electricity Production S2 (High) [PJ/a]	0	0	0	0	0	0	1.3374	2.02	2.8019	3.6947	4.6656	
Decentralized Electricity Production S3 (Base) [PJ/a]	0	0	0	0	0	0	0.8039	1.202	1.6513	2.1564	2.6953	
Decentralized Electricity Production S3 (Low) [PJ/a]	0	0	0	0	0	0	0.7297	1.0796	1.4674	1.8954	2.3443	
Decentralized Electricity Production S3 (High) [PJ/a]	0	0	0	0	0	0	0.8694	1.3129	1.8212	2.4016	3.0326	

Table 0.3. Demand inputs from 2010 to 2020

Year	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Final Electricity Demand (Base) [PJ/a]	33.7464	35.7012	37.7712	39.96	42.2784	44.73	47.3256	50.0724	52.9812	55.5012
Final Electricity Demand (Low) [PJ/a]	29.0376	30.3912	31.806	33.2892	34.8408	36.468	38.1708	39.9528	41.8212	43.3368
Final Electricity Demand (High) [PJ/a]	38.3436	40.9716	43.7796	46.782	49.9932	53.424	57.0924	61.0164	65.214	69.012
Exports (Base) [PJ/a]	3.3746	3.5701	3.7771	3.996	4.2278	4.473	4.7326	5.0072	5.2981	5.5501
Exports (Low) [PJ/a]	2.9038	3.0391	3.1806	3.3289	3.4841	3.6468	3.8171	3.9953	4.1821	4.3337
Exports (High)	3.8344	4.0972	4.378	4.6782	4.9993	5.3424	5.7092	6.1016	6.5214	6.9012

[PJ/a]											
Decentralized Electricity Production S2 (Base) [PJ/a]	5.062	6.0692	7.1766	8.3916	9.724	11.1827	12.7778	14.521	16.4243	18.3154	
Decentralized Electricity Production S2 (Low) [PJ/a]	4.3556	5.1664	6.0433	6.9908	8.0132	9.117	10.3061	11.5862	12.9647	14.301	
Decentralized Electricity Production S2 (High) [PJ/a]	5.7517	6.9653	8.3182	9.8244	11.4984	13.356	15.4148	17.6947	20.2162	22.774	
Decentralized Electricity Production S3 (Base) [PJ/a]	3.2904	3.9449	4.6649	5.4547	6.3205	7.2688	8.3056	9.4385	10.6758	11.9048	
Decentralized Electricity Production S3 (Low) [PJ/a]	2.831	3.3581	3.928	4.5439	5.2088	5.926	6.6989	7.5312	8.4269	9.2959	
Decentralized Electricity Production S3 (High) [PJ/a]	3.7386	4.5274	5.4068	6.3857	7.474	8.6814	10.0199	11.5016	13.1407	14.8032	

Table 0.4. Demand inputs from 2021 to 2030

Year	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
Final Electricity Demand (Base) [PJ/a]	58.14	60.9048	63.8064	66.8448	70.0272	73.3284	76.7844	80.406	77.1358	79.4045
Final Electricity Demand (Low) [PJ/a]	44.9064	46.5372	48.222	49.9716	51.786	53.6796	55.6452	57.6828	57.0352	58.5277
Final Electricity Demand (High) [PJ/a]	73.0296	77.2884	81.7956	86.5692	91.6236	96.9264	102.5388	108.4824	100.6931	103.9028
Exports (Base) [PJ/a]	5.814	6.0905	6.3806	6.6845	7.0027	7.3328	7.6784	8.0406	7.7137	7.9405
Exports (Low) [PJ/a]	4.4906	4.6537	4.8222	4.9972	5.1786	5.368	5.5645	5.7683	5.7035	5.8528
Exports (High) [PJ/a]	7.303	7.7288	8.1796	8.6569	9.1624	9.6926	10.2539	10.8482	10.0693	10.3903
Decentralized Electricity Production S2 (Base) [PJ/a]	20.349	22.5349	24.8846	27.4064	30.1118	32.998	36.0886	39.3991	39.3394	42.0844
Decentralized Electricity Production S2 (Low) [PJ/a]	15.7172	17.2188	18.8068	20.4883	22.2682	24.156	26.1533	28.2647	29.088	31.0198
Decentralized Electricity	25.5604	28.5966	31.9003	35.4935	39.398	43.6169	48.1932	53.1565	51.3533	55.0685

Production S2 (High) [PJ/a]										
Decentralized Electricity Production S3 (Base) [PJ/a]	13.2268	14.6477	16.1752	17.8142	19.5728	21.4488	23.4576	25.6093	25.5704	27.355
Decentralized Electricity Production S3 (Low) [PJ/a]	10.2161	11.192	12.2242	13.3175	14.4742	15.7014	16.9996	18.3719	18.9072	20.1629
Decentralized Electricity Production S3 (High) [PJ/a]	16.6144	18.5879	20.7353	23.0706	25.609	28.3511	31.3258	34.5517	33.3796	35.7944

Table 0.5. Demand inputs from 2031 to 2040

Year	2041	2042	2043	2044	2045	2046	2047	2048	2049	2050
Final Electricity Demand (Base) [PJ/a]	81.6736	83.9426	86.2114	88.4804	90.7495	93.0182	95.2873	97.5564	99.8251	102.0942
Final Electricity Demand (Low) [PJ/a]	60.0203	61.5128	63.005	64.4976	65.9902	67.4824	68.9749	70.4675	71.9597	73.4522
Final Electricity Demand (High) [PJ/a]	107.1122	110.322	113.5318	116.7415	119.9513	123.161	126.3708	129.5806	132.7903	135.9997
Exports (Base) [PJ/a]	8.1673	8.3941	8.6213	8.8481	9.0749	9.3017	9.5288	9.7556	9.9824	10.2096
Exports (Low) [PJ/a]	6.002	6.1513	6.3005	6.4498	6.599	6.7482	6.8975	7.0468	7.196	7.3452
Exports (High) [PJ/a]	10.7112	11.0322	11.3532	11.6741	11.9951	12.3161	12.6371	12.9581	13.279	13.6
Decentralized Electricity Production S2 (Base) [PJ/a]	44.9204	47.8472	50.8648	53.973	57.1723	60.462	63.8424	67.3139	70.8757	74.5286
Decentralized Electricity Production S2 (Low) [PJ/a]	33.0113	35.0622	37.1729	39.3437	41.5739	43.8635	46.2132	48.6223	51.0916	53.6202
Decentralized Electricity Production S2 (High) [PJ/a]	58.9118	62.8837	66.9838	71.2123	75.5694	80.0546	84.6684	89.4103	94.2811	99.2801
Decentralized Electricity Production S3 (Base) [PJ/a]	29.1982	31.1008	33.062	35.0824	37.1621	39.3005	41.4976	43.754	46.0692	48.4438
Decentralized Electricity Production S3	21.4571	22.7905	24.1625	25.5733	27.023	28.5113	30.0384	31.6048	33.2093	34.853

(Low) [PJ/a]										
Decentralized Electricity Production S3 (High) [PJ/a]	38.2928	40.8744	43.5395	46.2881	49.1202	52.0355	55.0343	58.117	61.2828	64.5318

Table 0.6: Demand inputs from 2041 to 2050

Technology	Variable Cost [K\$/PJ]	Fixed Cost [K\$/GW/y]	Capacity Factor [share]	Efficiency [share]	Capacity (Residual) [GW]	Firm Production [PJ/a]	First year	Phase Out Year
Electricity Imports	55556	-	-	1	-	0.72	2010	-
HFO PP	2750	125000	0.8	0.35	0.06	1.512	2010	2035
Biomass PP	1111	46000	0.75	0.35	0.037	0.8748	2011	2036
Diesel PP	2750	125000	0.75	0.35	0.1	2.3652	2010	2027
Coal PP1	1444	60000	0.8	0.4	0.2	5.0472	2013	2048
Coal PP2	1444	60000	0.8	0.4	0.4	10.0908	2024	-
Coal PP3	1444	60000	0.8	0.4	0.4	10.0908	2025	-
NG PP1	2750	125000	0.75	0.4	0.042	0.9936	2010	2019
NG PP2	2750	125000	0.75	0.4	0.12	2.8368	2010	2019
NG PP3	2750	125000	0.75	0.4	0.04	0.9468	2010	2022
NG PP4	2750	125000	0.75	0.4	0.1	2.3652	2010	2033
NG PP5 (CC)	861	36000	0.75	0.55	0.045	1.0656	2010	2034
NG PP6	2750	125000	0.75	0.4	0.1	2.3652	2011	2036
NG PP7	2750	125000	0.75	0.4	0.24	5.6772	2013	2038
NG PP8 (CC)	861	36000	0.75	0.55	0.3	7.0956	2017	2042
Hydro 1	-	10000	0.63	-	0.08	0.7848	2010	-
Hydro 2	-	10000	0.51	-	0.204	2.3256	2010	-
Hydro 3	-	10000	0.44	-	0.021	0.1944	2010	2045
Hydro 4	-	10000	0.42	-	0.18	1.7676	2010	-
Hydro 5	-	10000	0.49	-	0.068	0.6912	2010	-
Hydro 6	-	10000	0.48	-	0.008	0.0756	2010	2047
Hydro 7	-	10000	0.61	-	0.358	4.9572	2010	-
Hydro 8	-	10000	0.8	-	0.063	1.5084	2016	-
Hydro 9	-	10000	0.87	-	0.053	1.4544	2016	-
Hydro 10	-	10000	0.51	-	0.034	0.3852	2016	-
Hydro 11	-	10000	0.57	-	0.157	1.8432	2019	-
Hydro 12	-	10000	0.56	-	0.149	1.7784	2020	-
Hydro 13	-	10000	0.64	-	0.118	1.8684	2018	-
Hydro 14	-	10000	0.76	-	0.144	2.5128	2018	-
Hydro 15	-	10000	0.67	-	0.145	2.3652	2020	-
Hydro 16	-	10000	0.76	-	0.222	3.5568	2018	-
Hydro 17	-	10000	0.62	-	0.34	5.0148	2019	-
Hydro 18	-	10000	0.63	-	0.3	4.428	2019	-
Hydro 19	-	10000	0.63	-	0.6	8.856	2019	-
Hydro 20	-	10000	0.63	-	0.3	4.428	2019	-

Table A.7. Individual Power Plants

S1 Base	S1 Low	S1 High	S2 Base	S2 Low	S2 High	S3 Base	S3 Low	S3 High
0.5	0.35	0.5	0.15	0.1	0.15	0.25	0.15	0.25

Table A.8. Lower production limit coefficients

Technology	Variable Cost [K\$/PJ]	Fixed Cost [K\$/GW/y]	Capital cost [K\$/GW]	Capacity factor [share]	Efficiency [share]	Plant life [years]
Biomass PPs	1111	46000	1256000	0.75	0.35	25
Coal PPs	1444	60000	2000000	0.8	0.4	35
NG PPs (CC)	861	36000	750000	0.75	0.55	25
Geothermal PPs	-	35000	2238000	0.8	-	25
CSP	-	53000	3785000	0.40/0 (day/night)	-	25
Large Hydro	-	10000	2000000	0.6	-	80
Large Wind Farms	-	29000	1439000	0.25	-	25

Table 0.9. New technologies- Centralized Power System

Technology	Variable Cost [K\$/PJ]	Fixed Cost [K\$/GW/y]	Capital cost [K\$/GW]	Capacity factor [share]	Efficiency [share]	Plant life [years]
Small Biomass PPs	1111	46000	1256000	0.75	0.35	25
Diesel PPs	2750	125000	469000	0.75	0.35	25
Small Hydro	-	16000	2000000	0.45	-	25
Small Wind farms	-	40000	2370000	0.25	-	25
PV	-	27000	4000000	0.40/0 (day/night)	-	25

Table 0.10. New technologies- Decentralized Power System

	Variable Cost [K\$/PJ]	Fixed Cost [K\$/GW/y]	Capital cost [K\$/GW]	Plant life [years]
Backstop Technologies	99999	99999	99999	1

Table 0.11. Input Data for Backstop Technologies

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Coal [K\$/PJ]	2,130	2,253	2,319	2,292	2,380	2,420	2,519	2,573	2,622	2,670	2,747
HFO [K\$/PJ]	11,233	15,128	18,342	16,689	21,722	21,944	22,700	24,495	25,239	26,461	27,239
Diesel [K\$/PJ]	17,825	22,066	22,684	22,139	20,314	20,642	21,370	22,170	22,908	23,708	24,519
Natural Gas [K\$/PJ]	4,875	4,522	3,203	3,642	3,736	3,878	4,389	4,630	4,972	5,181	5,347
Biomass [K\$/PJ]	16,667	16,667	16,667	16,667	16,667	16,667	16,667	16,667	16,667	16,667	16,667

Table 0.12. Fuel prices from 2010 to 2020

Year	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Coal [K\$/PJ]	2,828	2,931	3,028	3,117	3,211	3,316	3,417	3,517	3,630	3,747
HFO [K\$/PJ]	28,114	29,036	30,142	31,408	32,575	33,698	34,789	35,897	37,075	38,372
Diesel [K\$/PJ]	25,403	26,319	27,302	28,297	29,375	30,427	31,519	32,650	33,836	35,025
Natural Gas [K\$/PJ]	5,564	5,886	6,205	6,469	6,658	6,936	7,128	7,392	7,656	7,911
Biomass [K\$/PJ]	16,667	16,667	16,667	16,667	16,667	16,667	16,667	16,667	16,667	16,667

Table 0.73. Fuel prices from 2021 to 2030

Year	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
Coal [K\$/PJ]	3,856	3,964	4,078	4,200	4,336	4,469	4,600	4,733	4,870	5,005
HFO [K\$/PJ]	39,908	41,278	42,642	44,025	45,580	46,897	48,469	50,375	52,264	53,997
Diesel [K\$/PJ]	36,245	37,475	38,828	40,272	41,836	43,481	45,156	46,606	48,358	50,073
Natural Gas [K\$/PJ]	8,233	8,533	8,900	9,417	9,989	10,700	11,383	12,108	12,528	13,094
Biomass [K\$/PJ]	16,667	16,667	16,667	16,667	16,667	16,667	16,667	16,667	16,667	16,667

Table 0.84. Fuel prices from 2031 to 2030

Year	2041	2042	2043	2044	2045	2046	2047	2048	2049	2050
Coal [K\$/PJ]	5,144	5,289	5,436	5,589	5,745	5,906	6,073	6,242	6,417	6,597
HFO [K\$/PJ]	56,427	58,967	61,620	64,392	67,292	70,319	73,483	76,789	80,245	83,856
Diesel [K\$/PJ]	51,525	53,019	54,556	56,139	57,767	59,441	61,167	62,939	64,764	66,642
Natural Gas [K\$/PJ]	13,580	14,083	14,603	15,144	15,703	16,286	16,889	17,514	18,161	18,833
Biomass [K\$/PJ]	16,667	16,667	16,667	16,667	16,667	16,667	16,667	16,667	16,667	16,667

Table 0.95. Fuel prices from 2041 to 2050

Geothermal annual potential: 7,2 PJ

Appendix B– Modeling outputs

Appendix B1 – Answer-OSeMOSYS

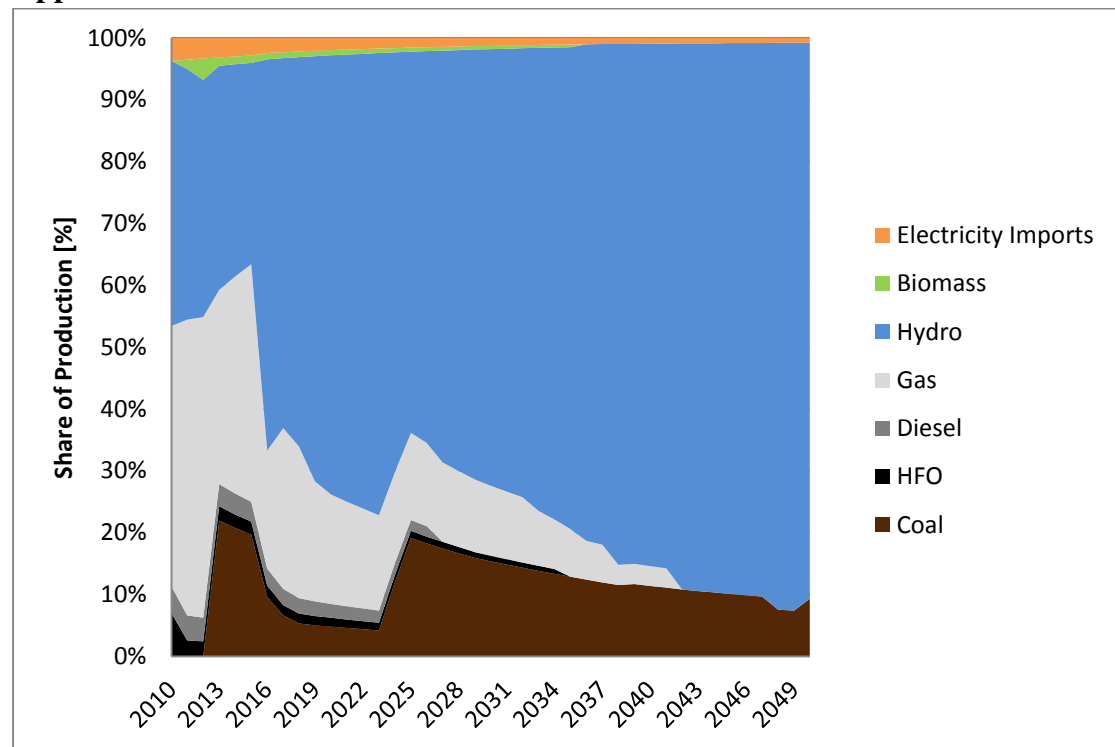


Figure A1. Answer-OSeMOSYS, Scenario 1(LOW)

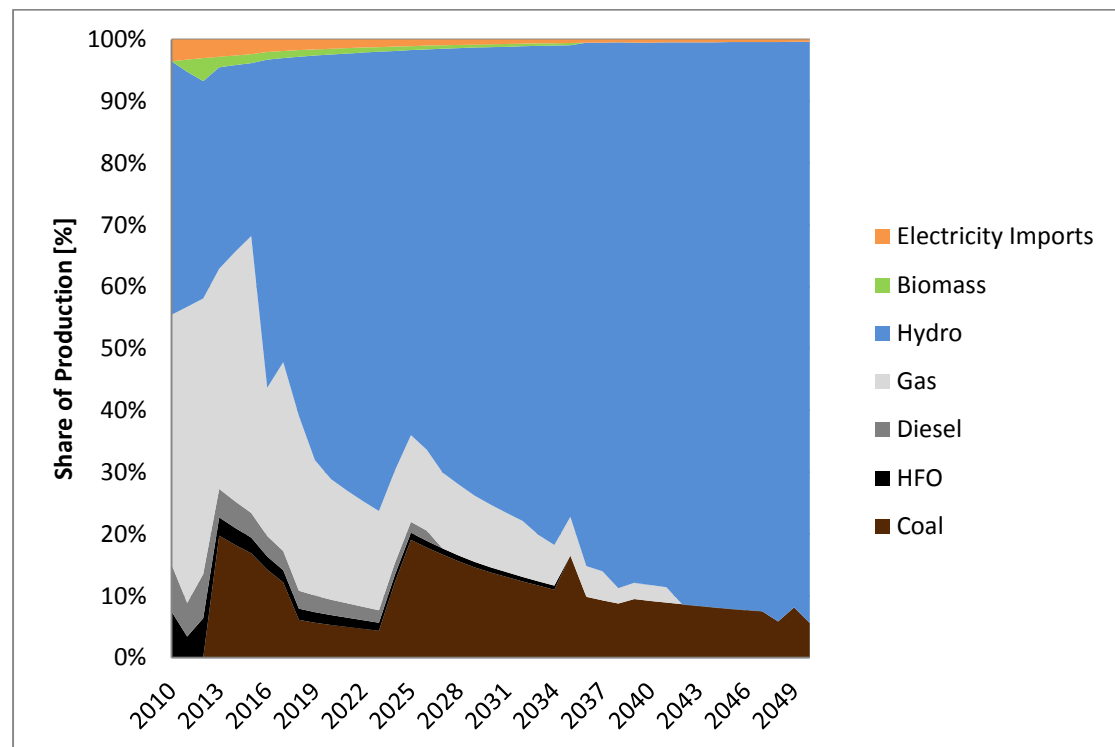


Figure A2. Answer-OSeMOSYS, Scenario 1(HIGH)

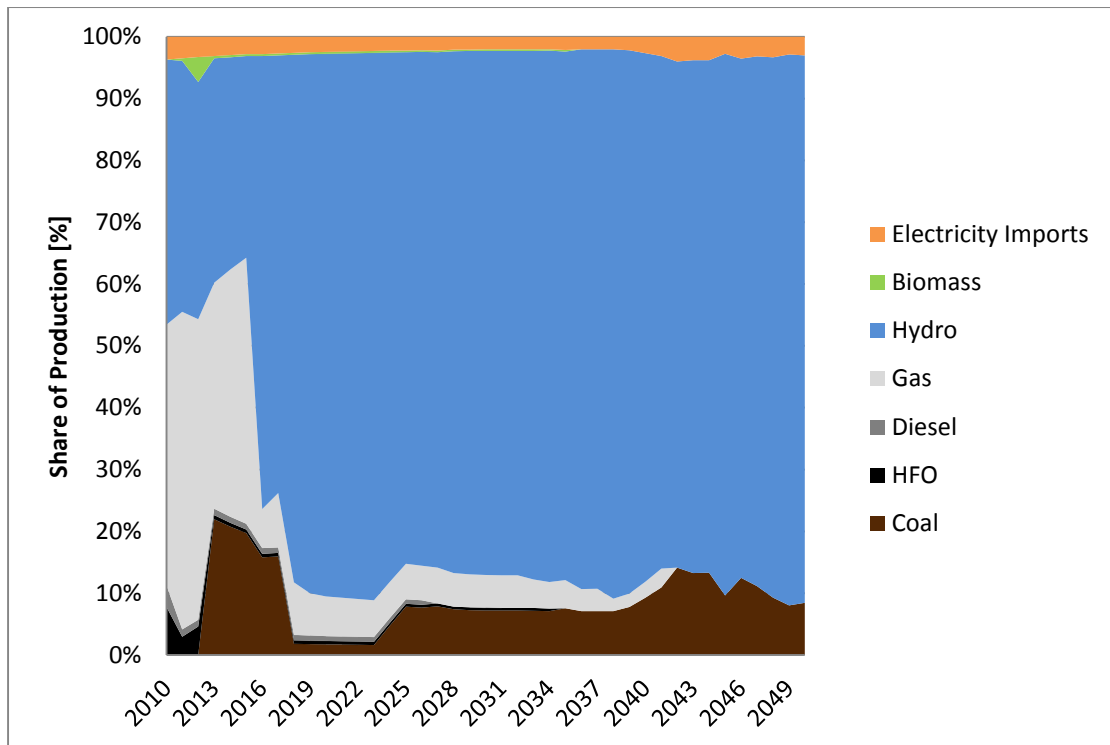


Figure A3. Answer-OSeMOSYS, Scenario 2(Low)

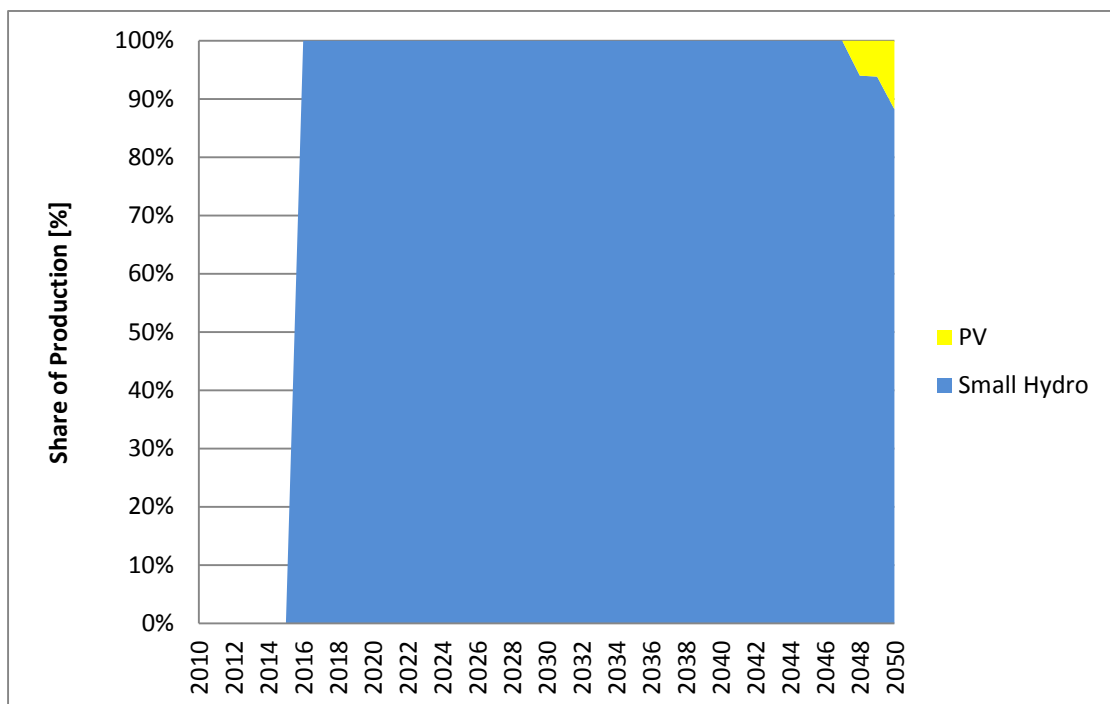


Figure A4. Answer-OSeMOSYS, Scenario 2(Low), Decentralized technologies

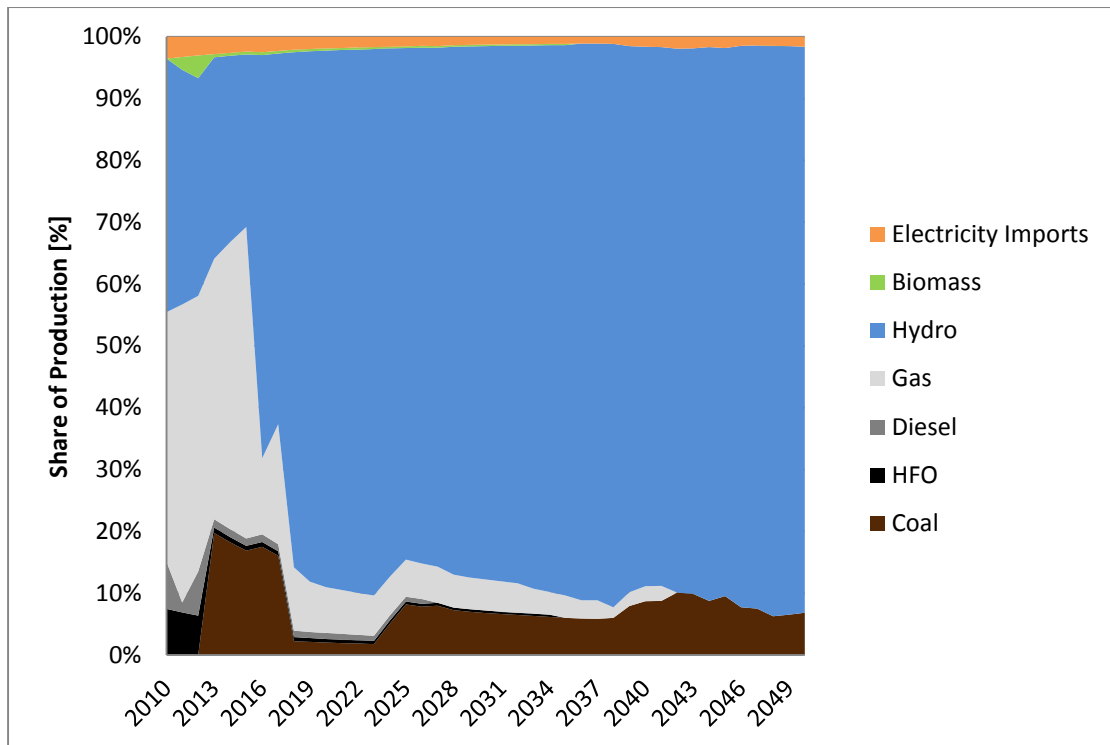


Figure A5. Answer-OSeMOSYS, Scenario 2(HIGH)

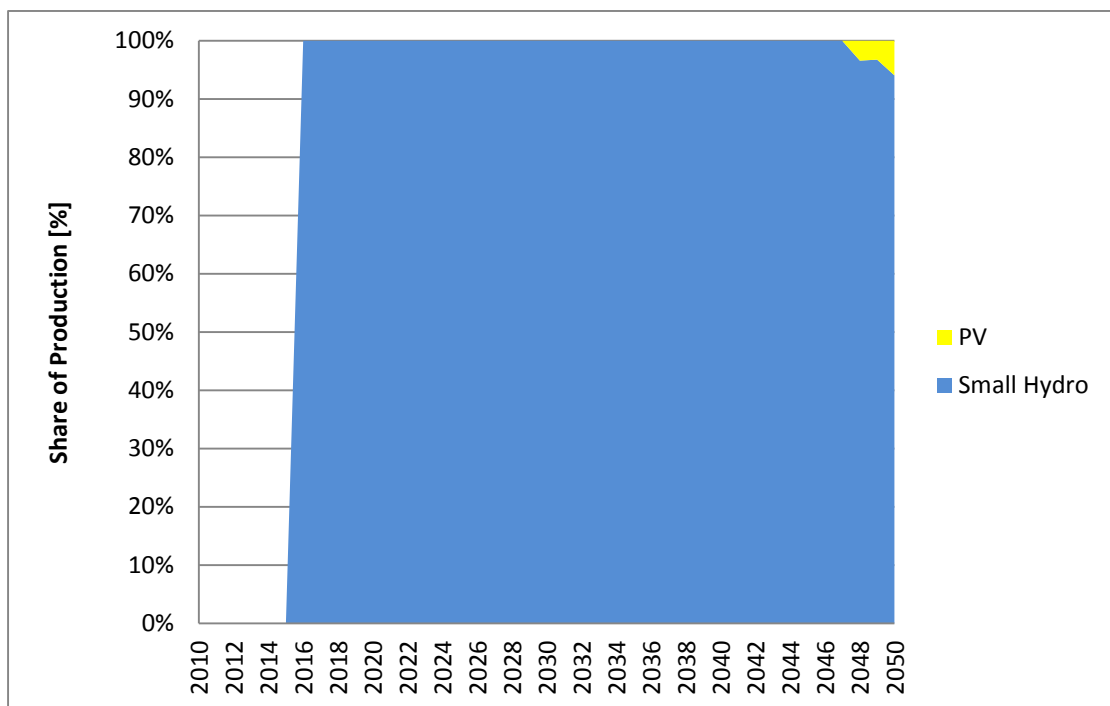


Figure A6. Answer-OSeMOSYS, Scenario 2(HIGH), Decentralized technologies

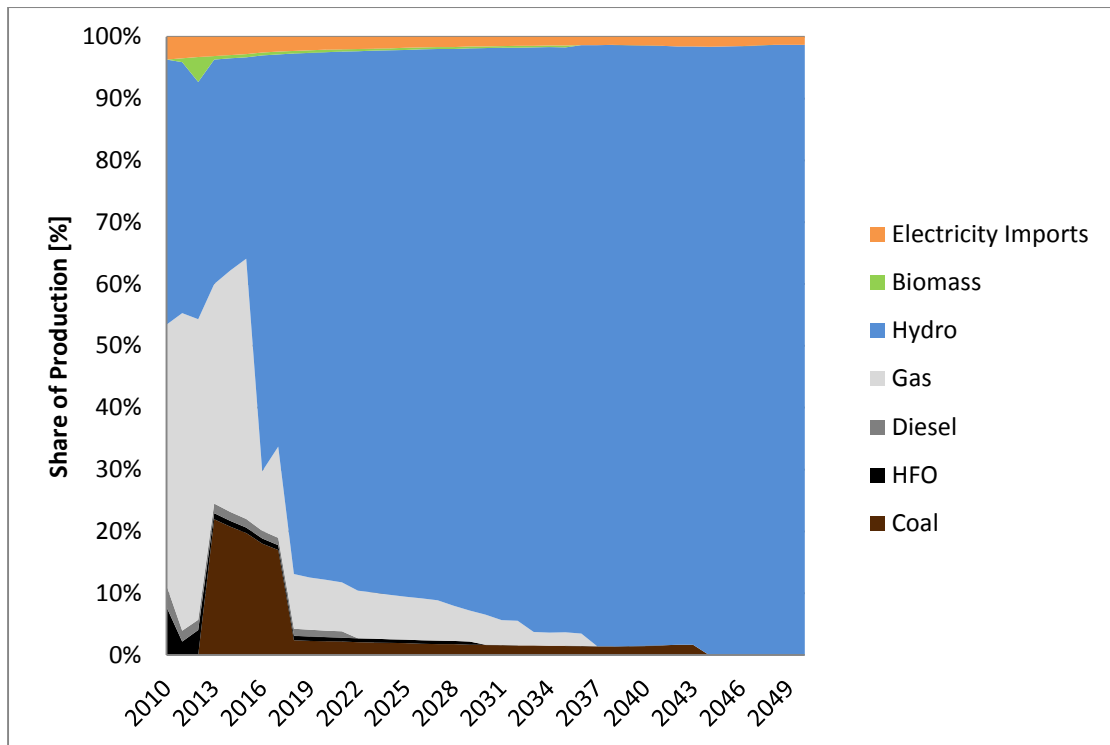


Figure A7. Answer-OSeMOSYS, Scenario 3(Low)

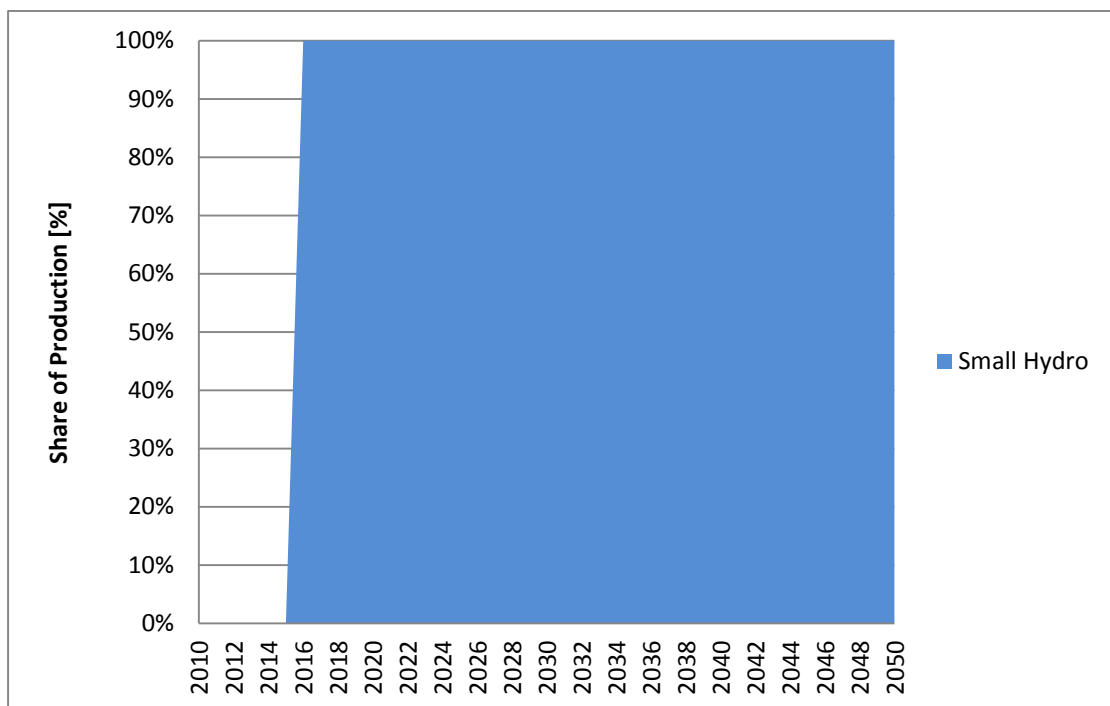


Figure A8. Answer-OSeMOSYS, Scenario 3(Low), Decentralized technologies

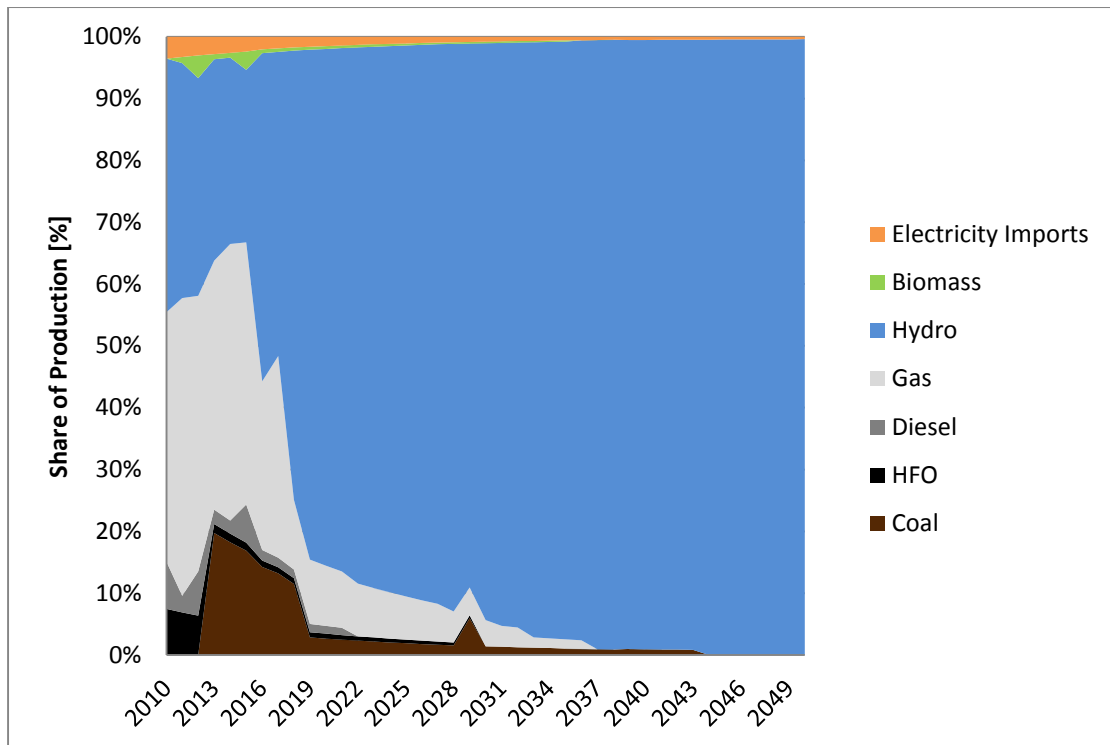


Figure A9. Answer-OSeMOSYS, Scenario 3(HIGH)

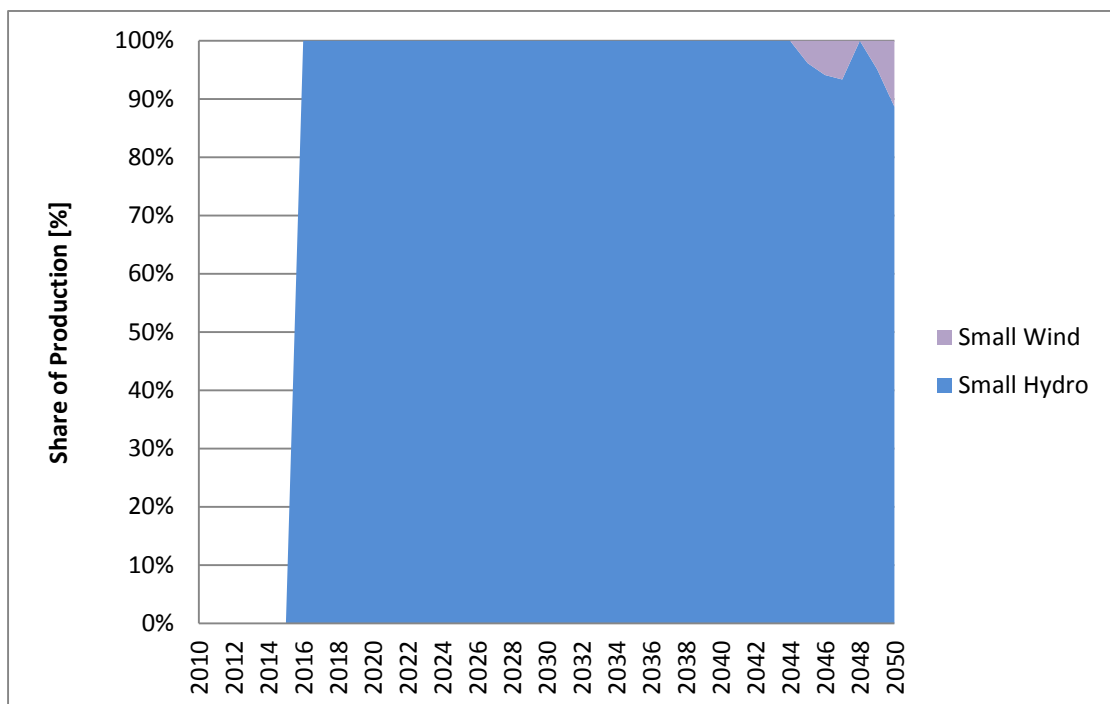


Figure A10. Answer-OSeMOSYS, Scenario 3(HIGH), Decentralized technologies

Appendix B2 – LEAP

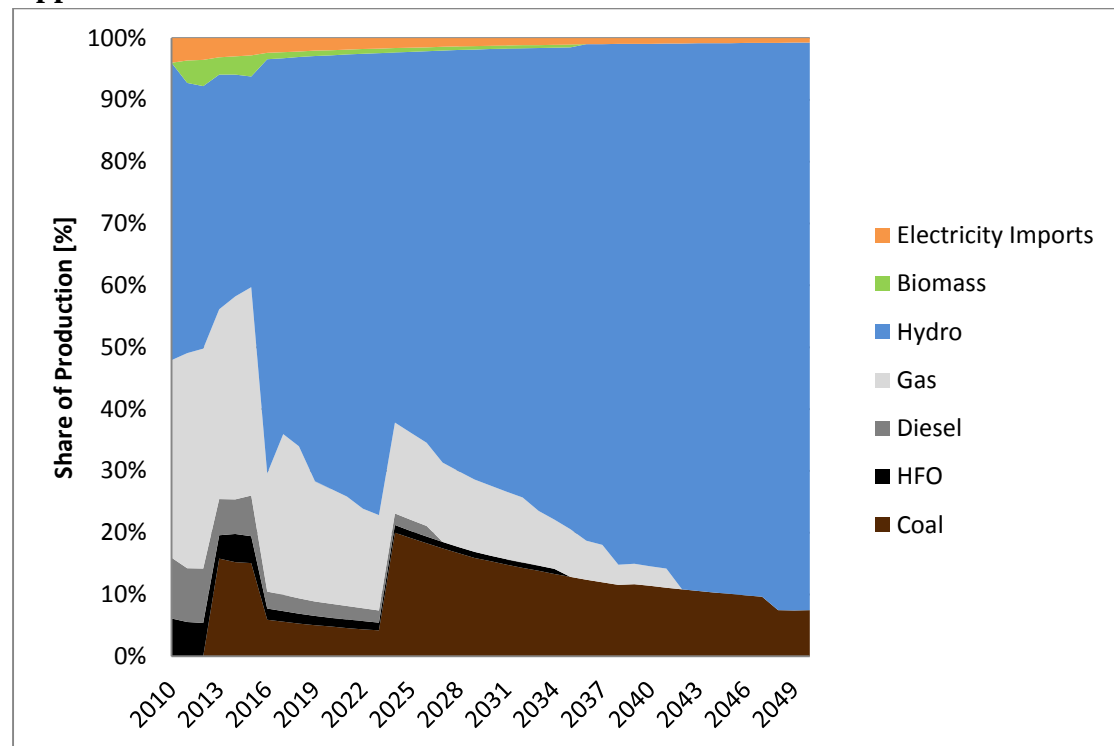


Figure A11. LEAP, Scenario 1(LOW)

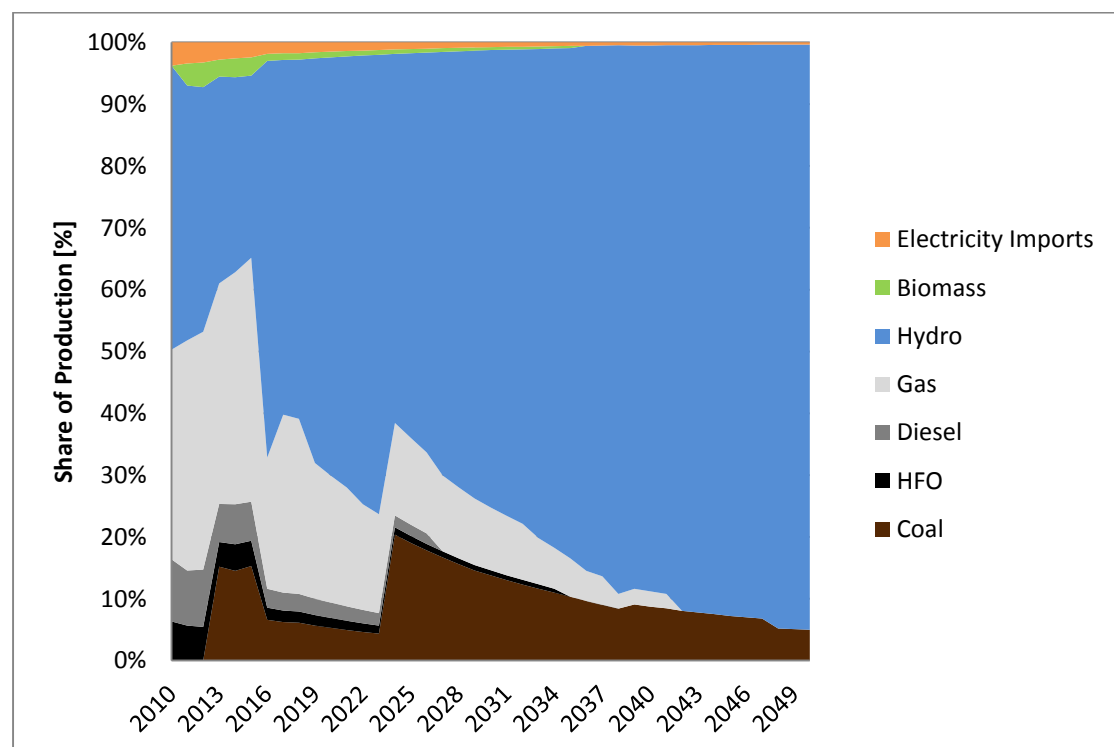


Figure A12. LEAP, Scenario 1(HIGH)

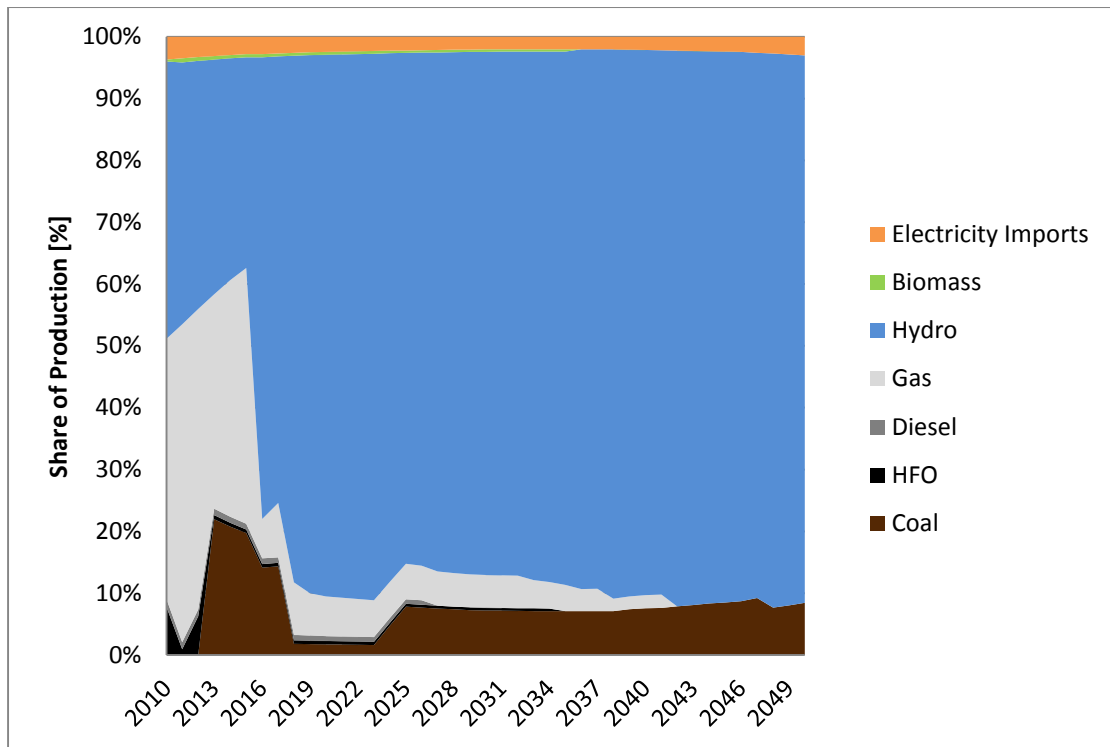


Figure A13. LEAP, Scenario 2(Low)

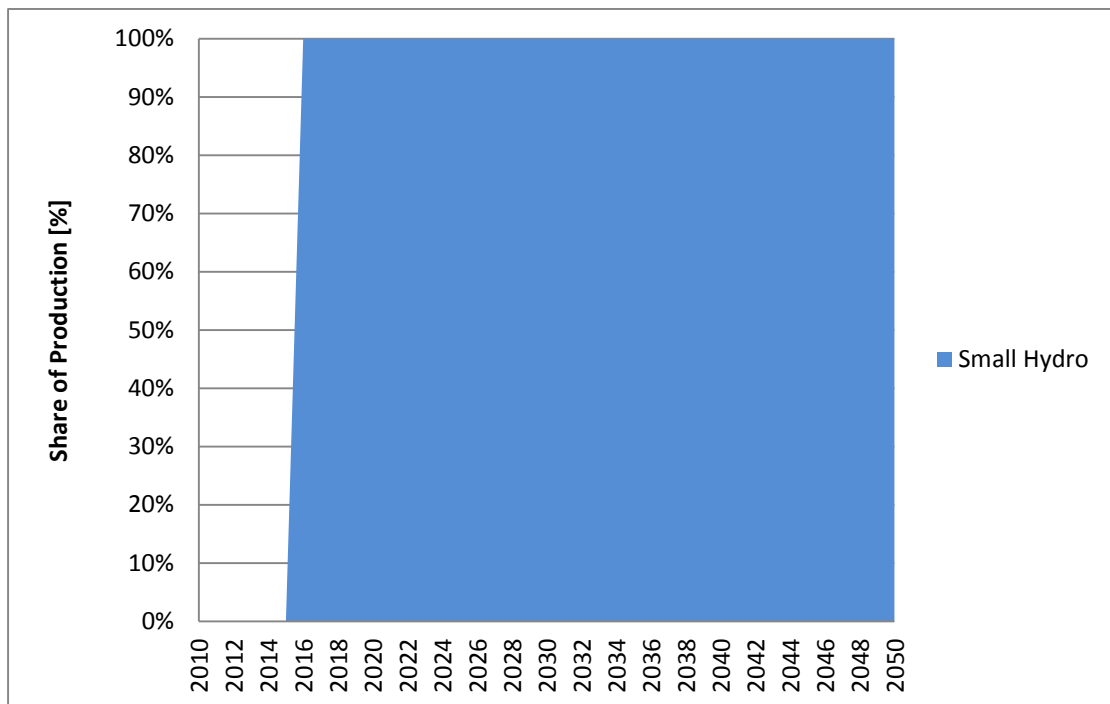


Figure A14. LEAP, Scenario 2(Low), Decentralized technologies

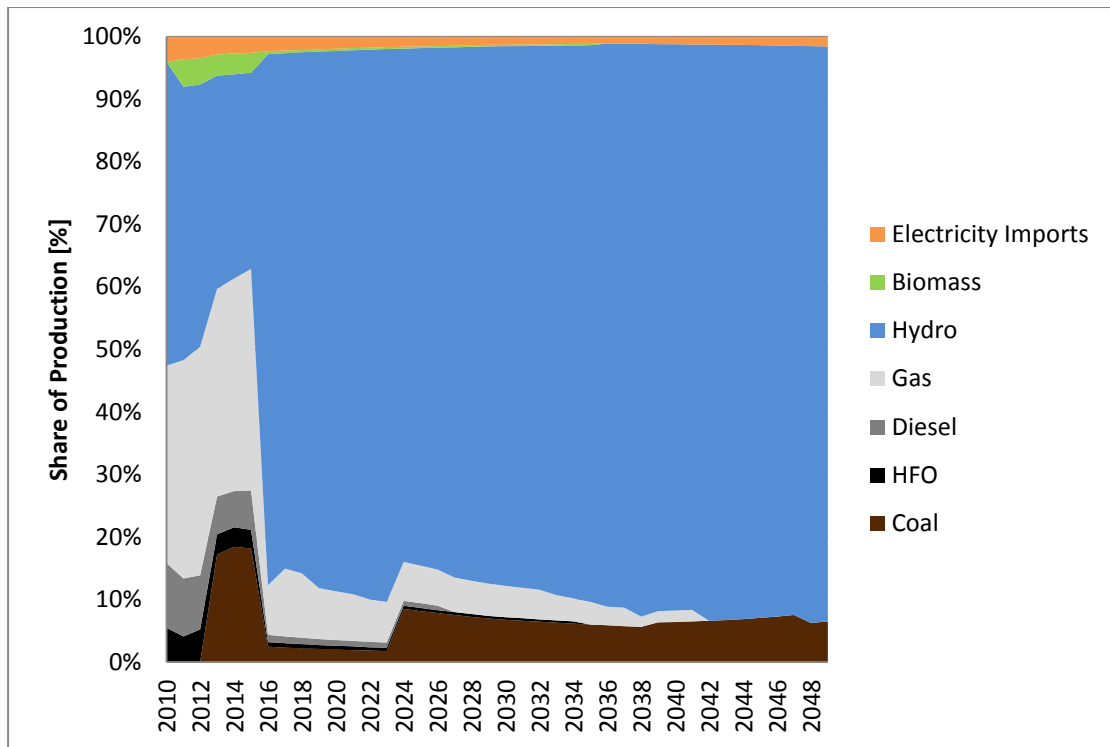


Figure A15. LEAP, Scenario 2(HIGH)

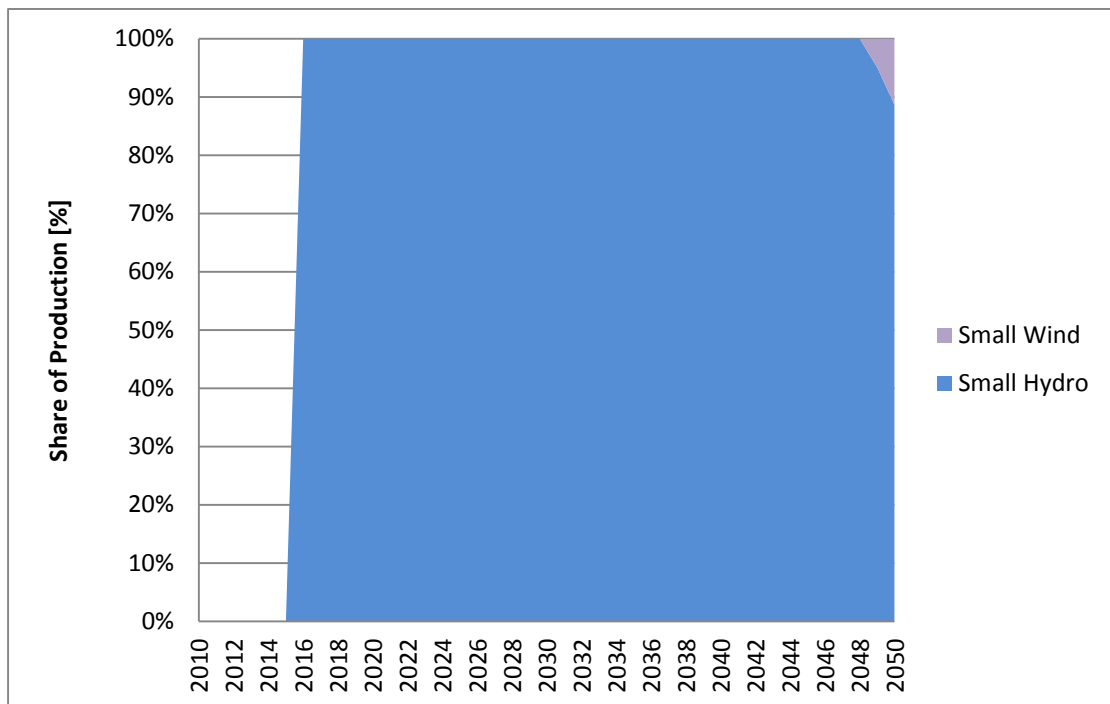


Figure A16. LEAP, Scenario 2(HIGH), Decentralized technologies

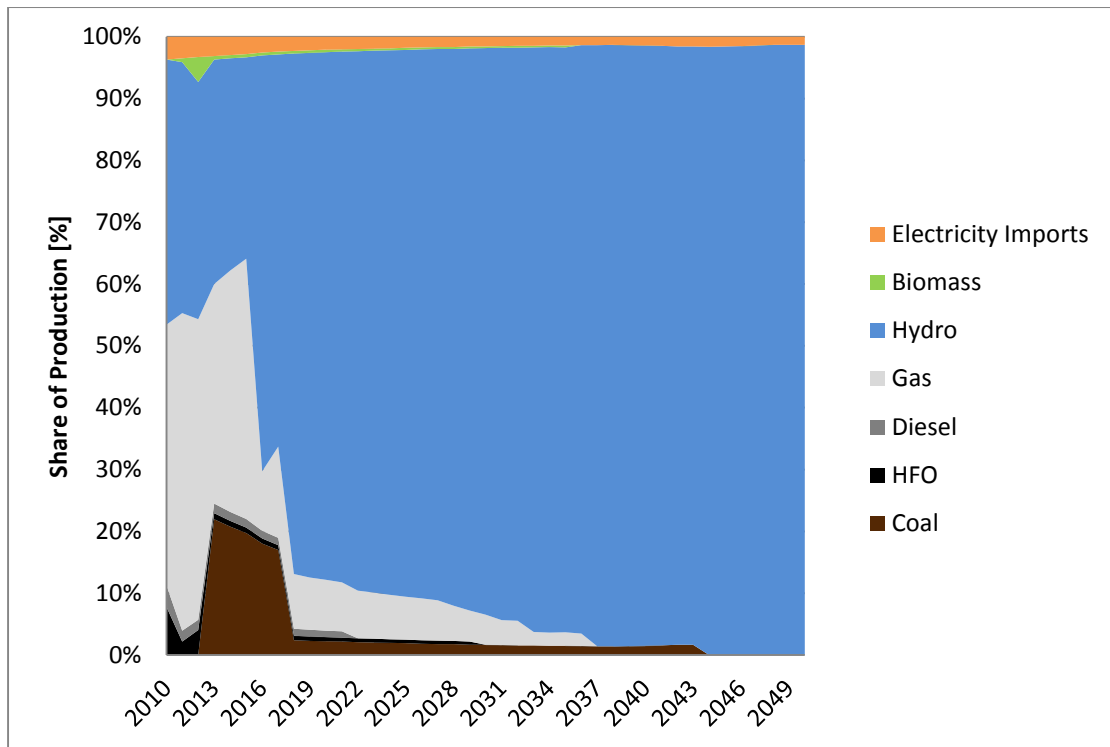


Figure A17. LEAP, Scenario 3(Low)

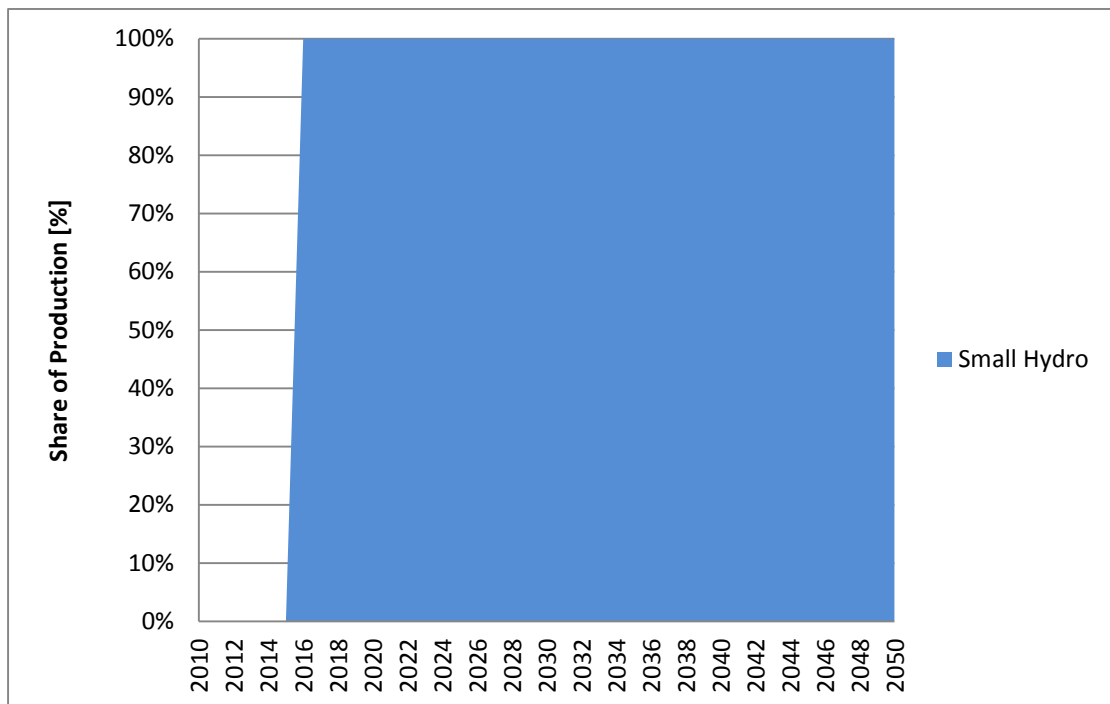


Figure A18. LEAP, Scenario 3(Low), Decentralized technologies

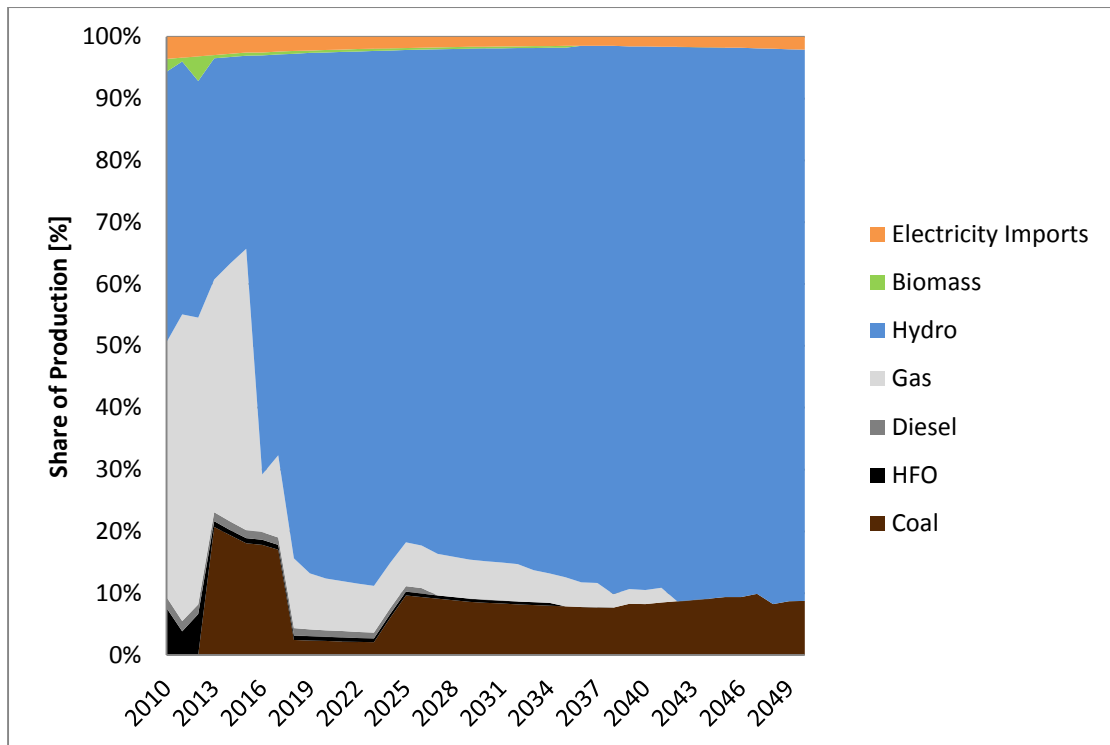


Figure A19. LEAP, Scenario 3(HIGH)

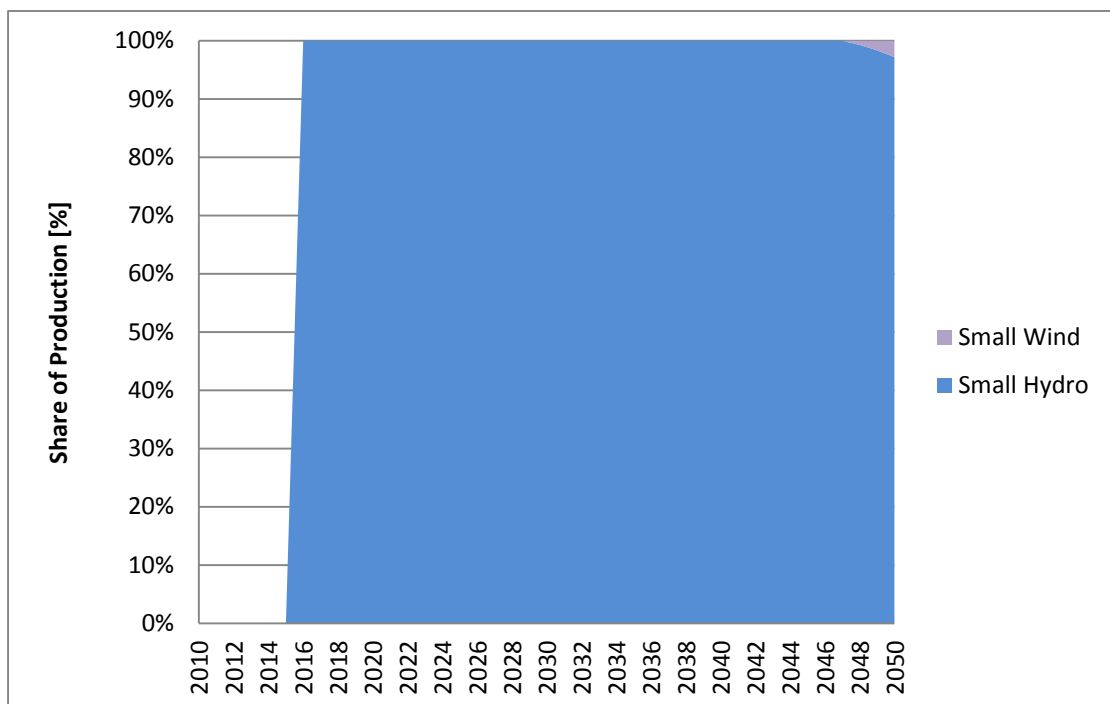


Figure A20. LEAP, Scenario 3(HIGH), Decentralized technologies

Appendix B3 – MESSAGE

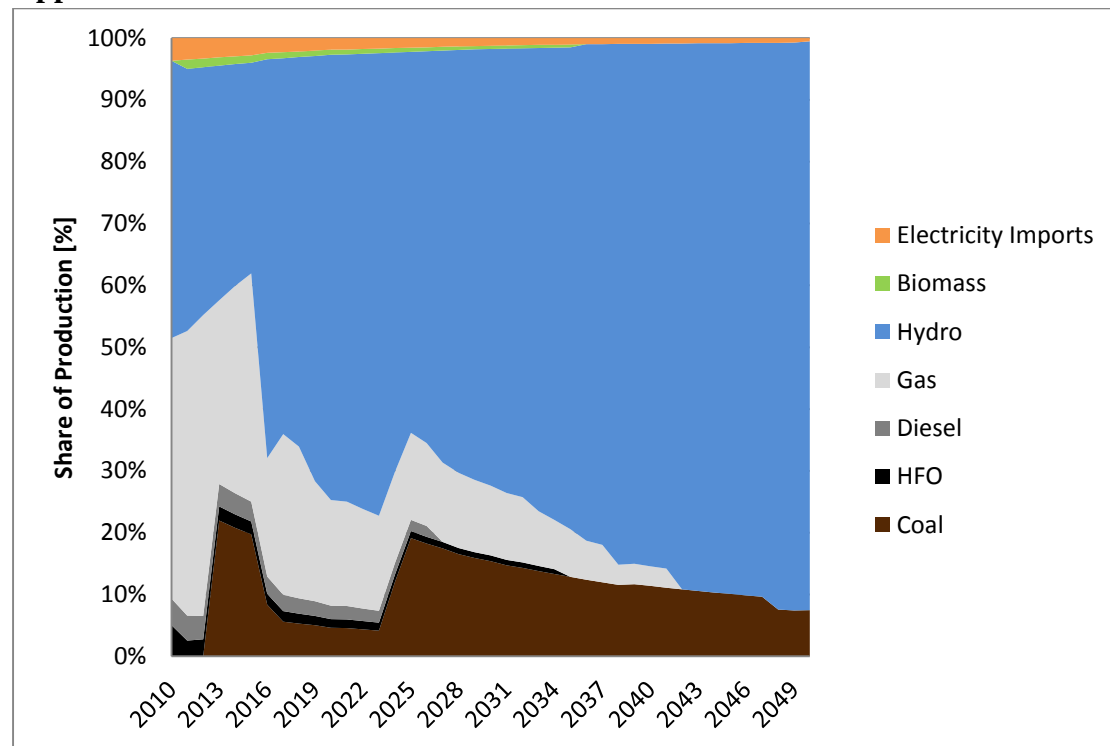


Figure A21. MESSAGE, Scenario 1(LOW)

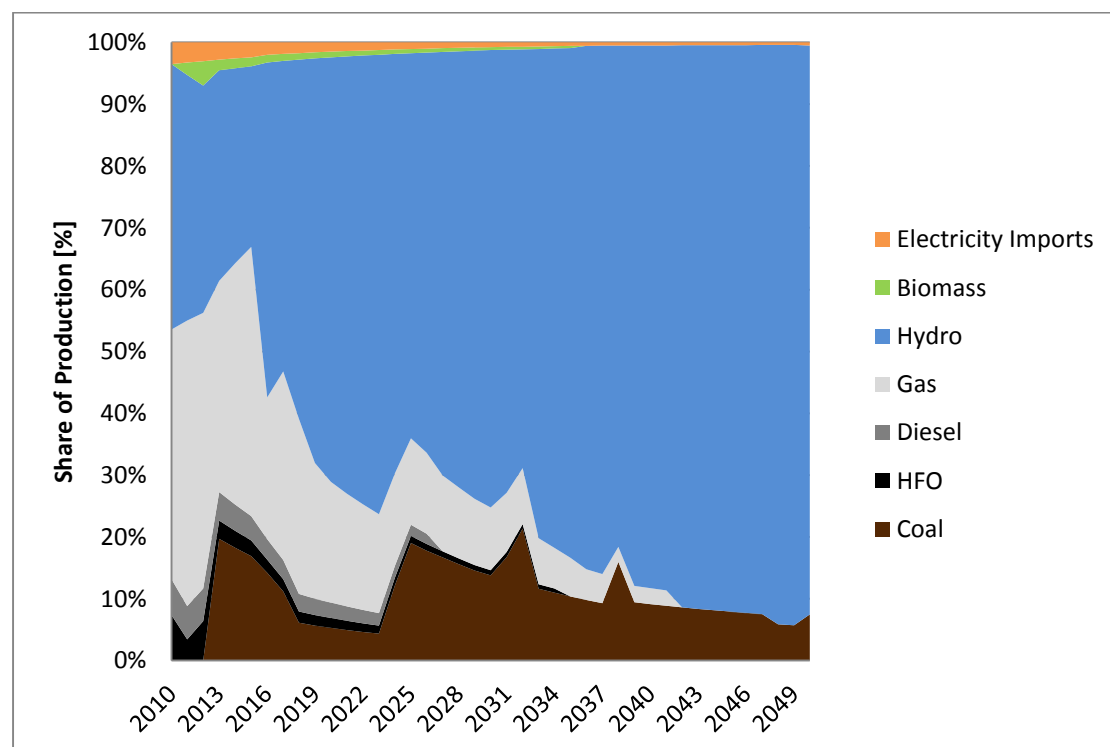


Figure A22. MESSAGE, Scenario 1(HIGH)

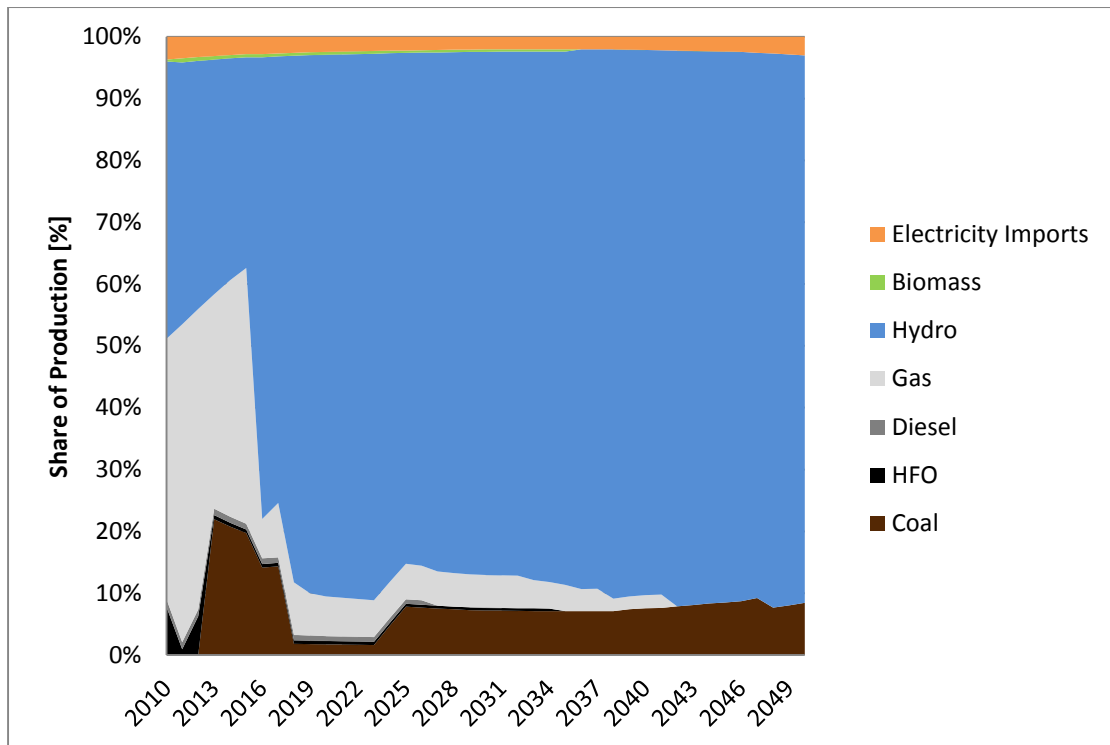


Figure A23. MESSAGE, Scenario 2(Low)

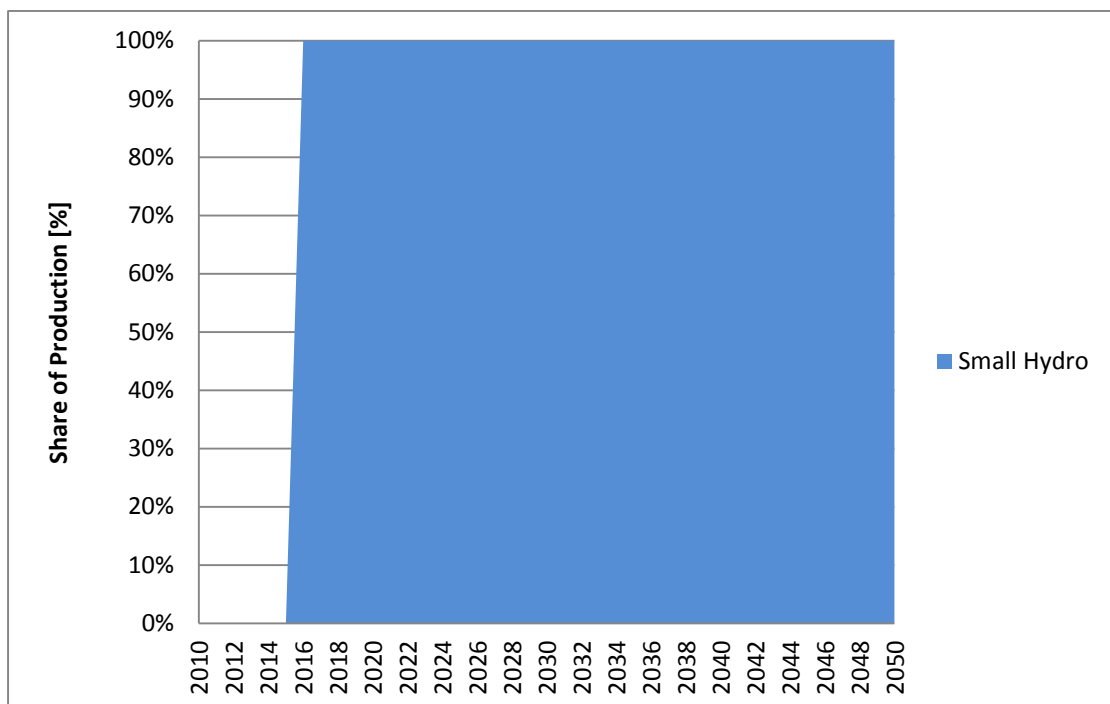


Figure A24. MESSAGE, Scenario 2(Low), Decentralized technologies

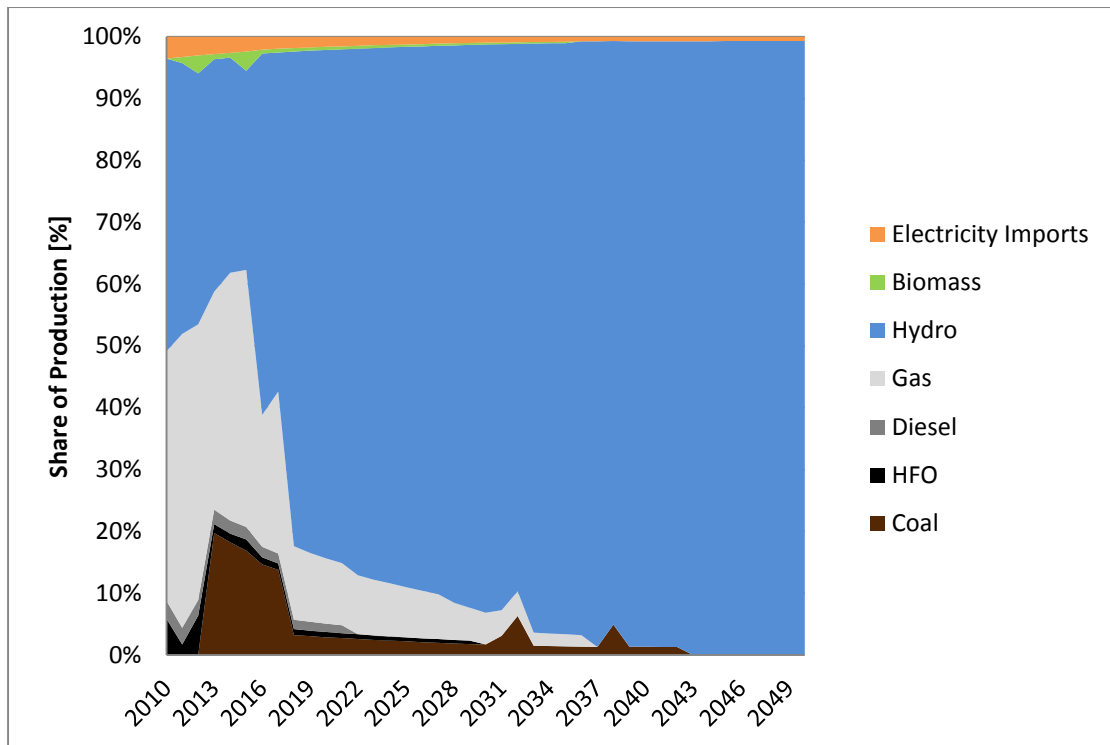


Figure A25. MESSAGE, Scenario 2(HIGH)

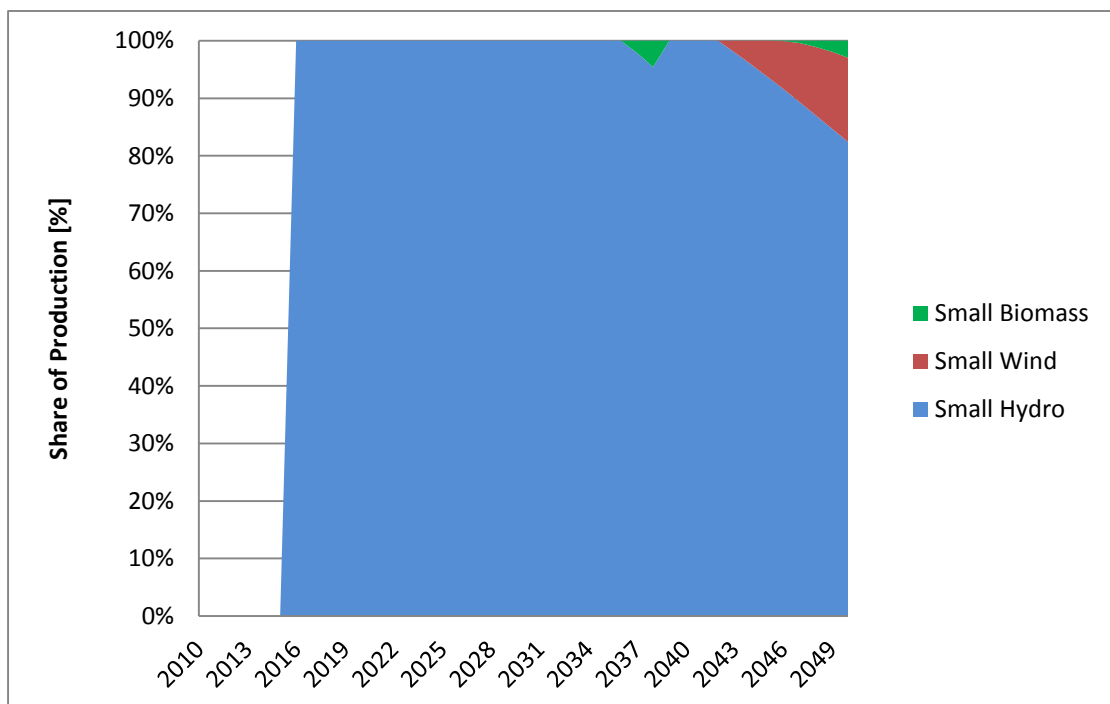


Figure A26. MESSAGE, Scenario 2(HIGH), Decentralized technologies

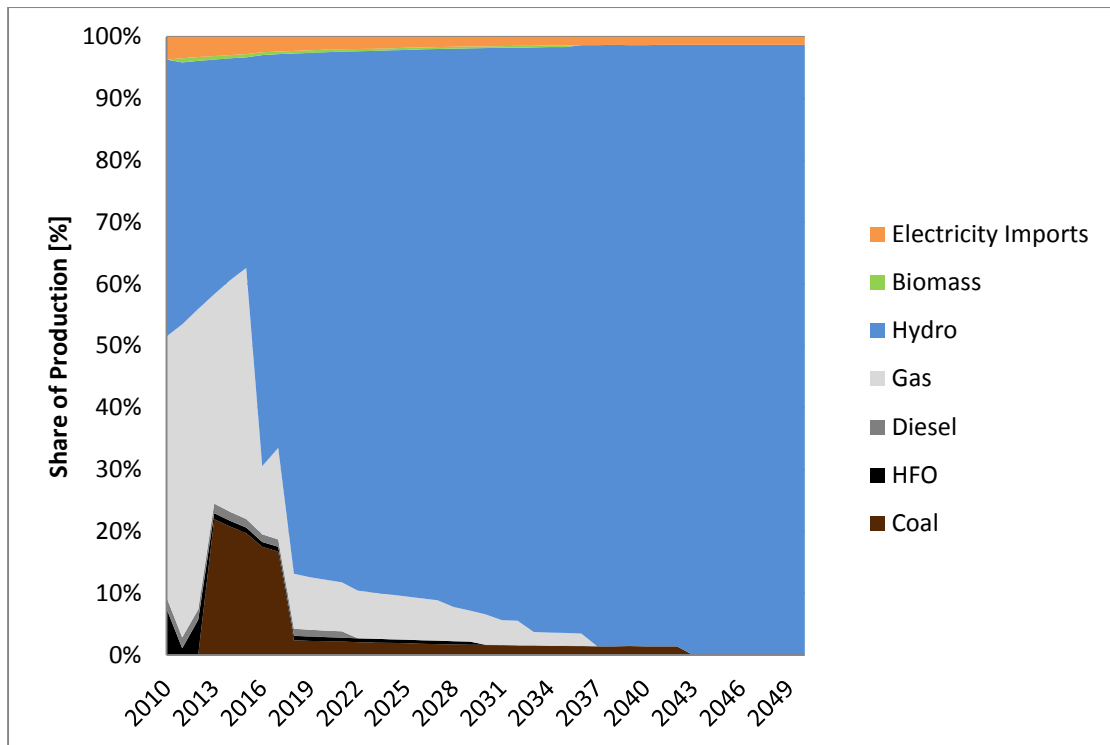


Figure A27. MESSAGE, Scenario 3(Low)

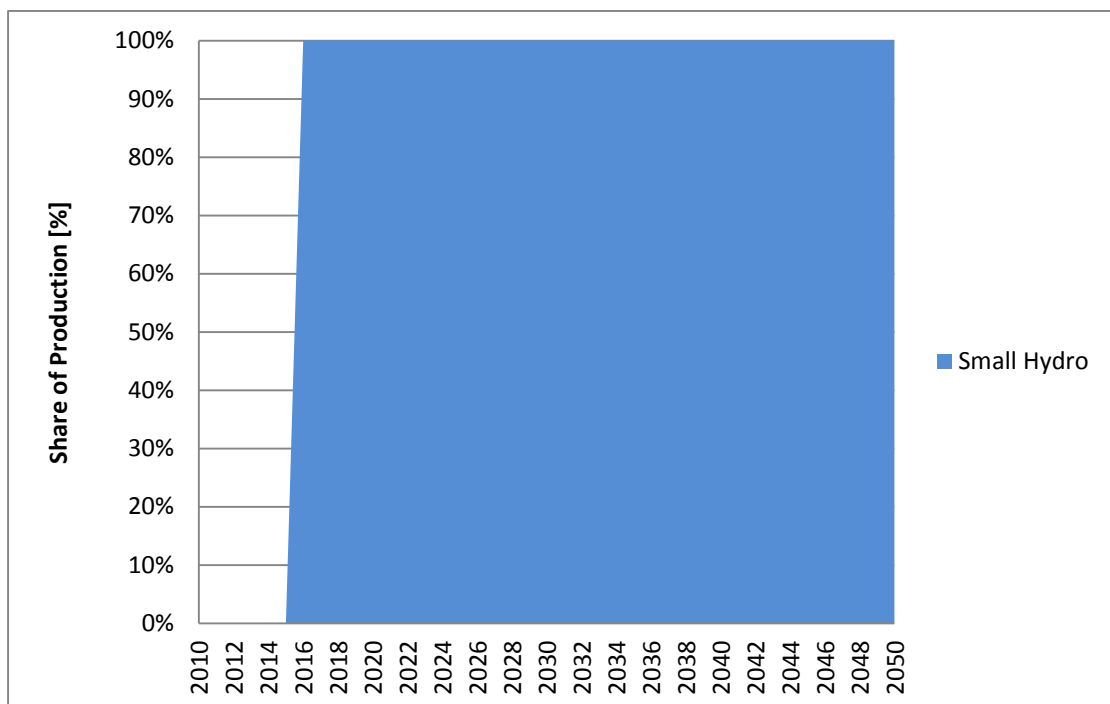


Figure A28. MESSAGE, Scenario 3(Low), Decentralized technologies

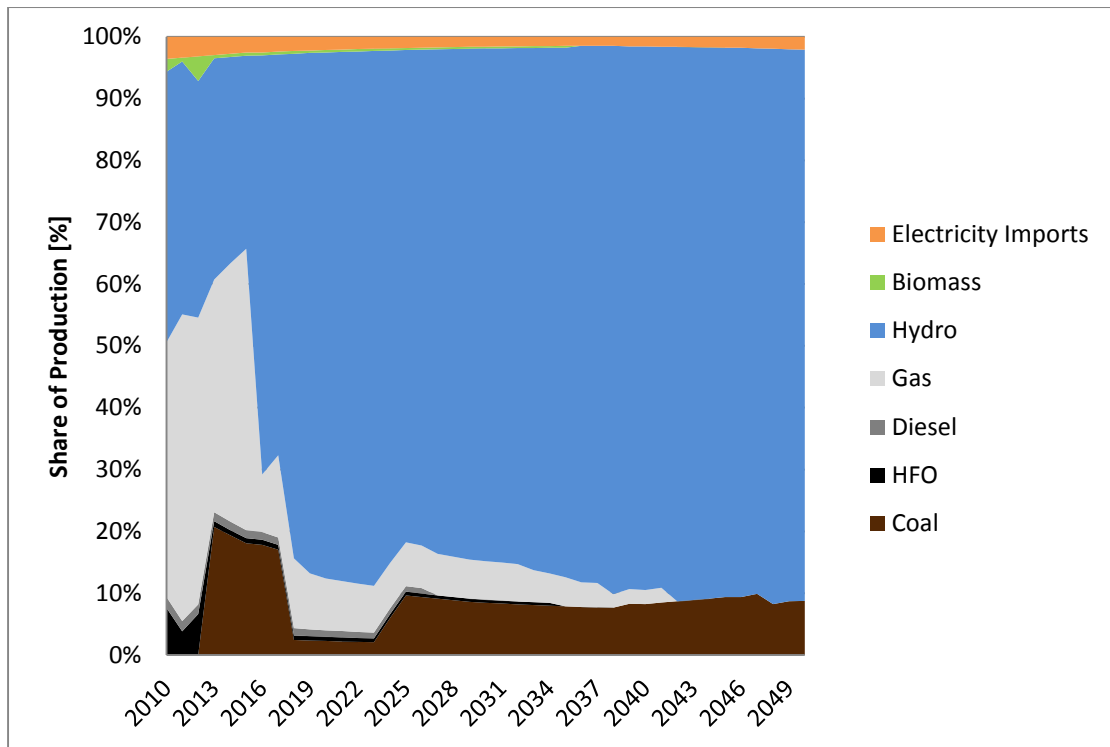


Figure 29. MESSAGE, Scenario 3(HIGH)

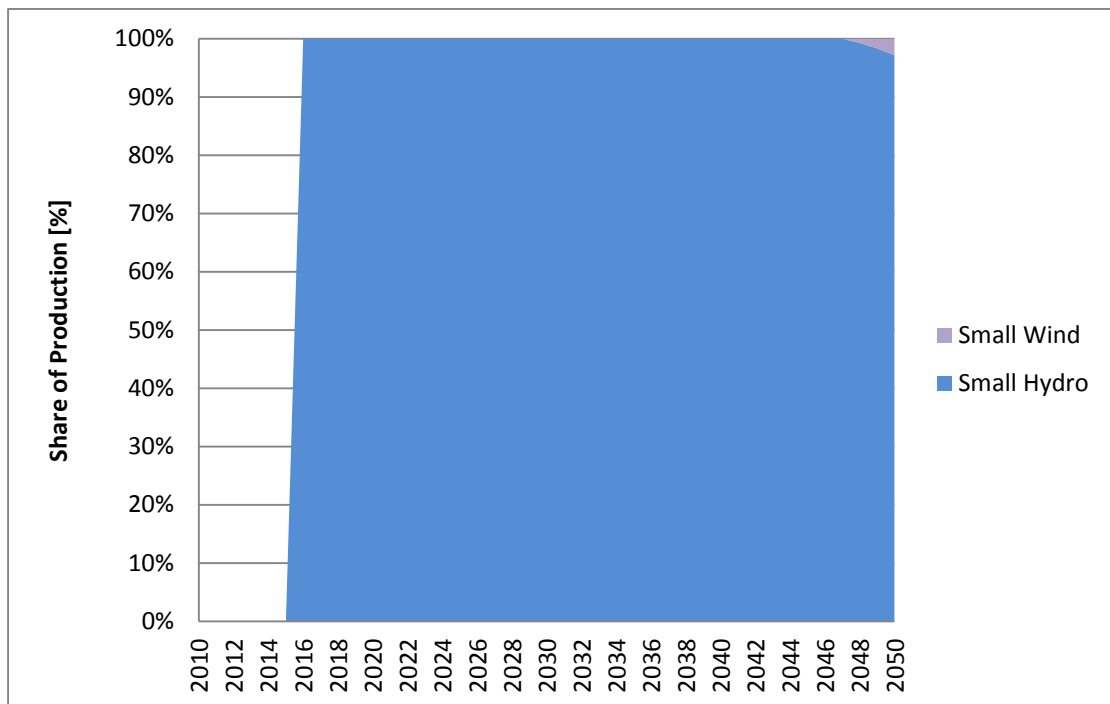


Figure A30. MESSAGE, Scenario 3(HIGH), Decentralized technologies

Appendix B4 – Sensitivity analysis

A sensitivity analysis –considering three scenarios- was done in Answer-OSeMOSYS based on Scenario 1 but the results were almost unaffected compared to the reference case. The parameters that subjected to change were the following:

1. Reduction of the annual capacity investment limit from 1 GW to 0.2 GW.
2. The above mentioned plus the hydro power plants that are planned to be built from 2016 onwards are not considered.
3. The two previous changes plus a 50% reduction in the cost of natural gas.